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Semester V

TYBCA

Cloud Computing Journal

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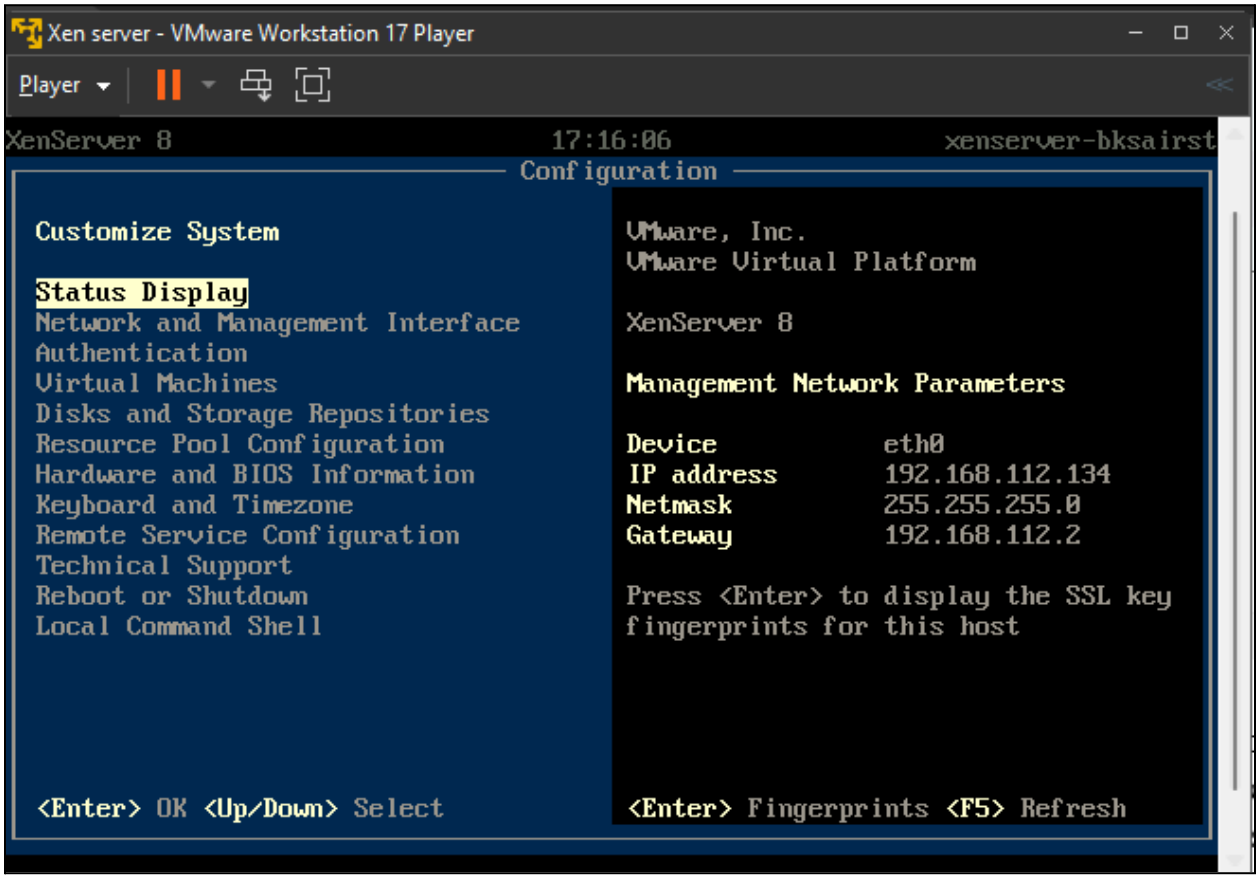
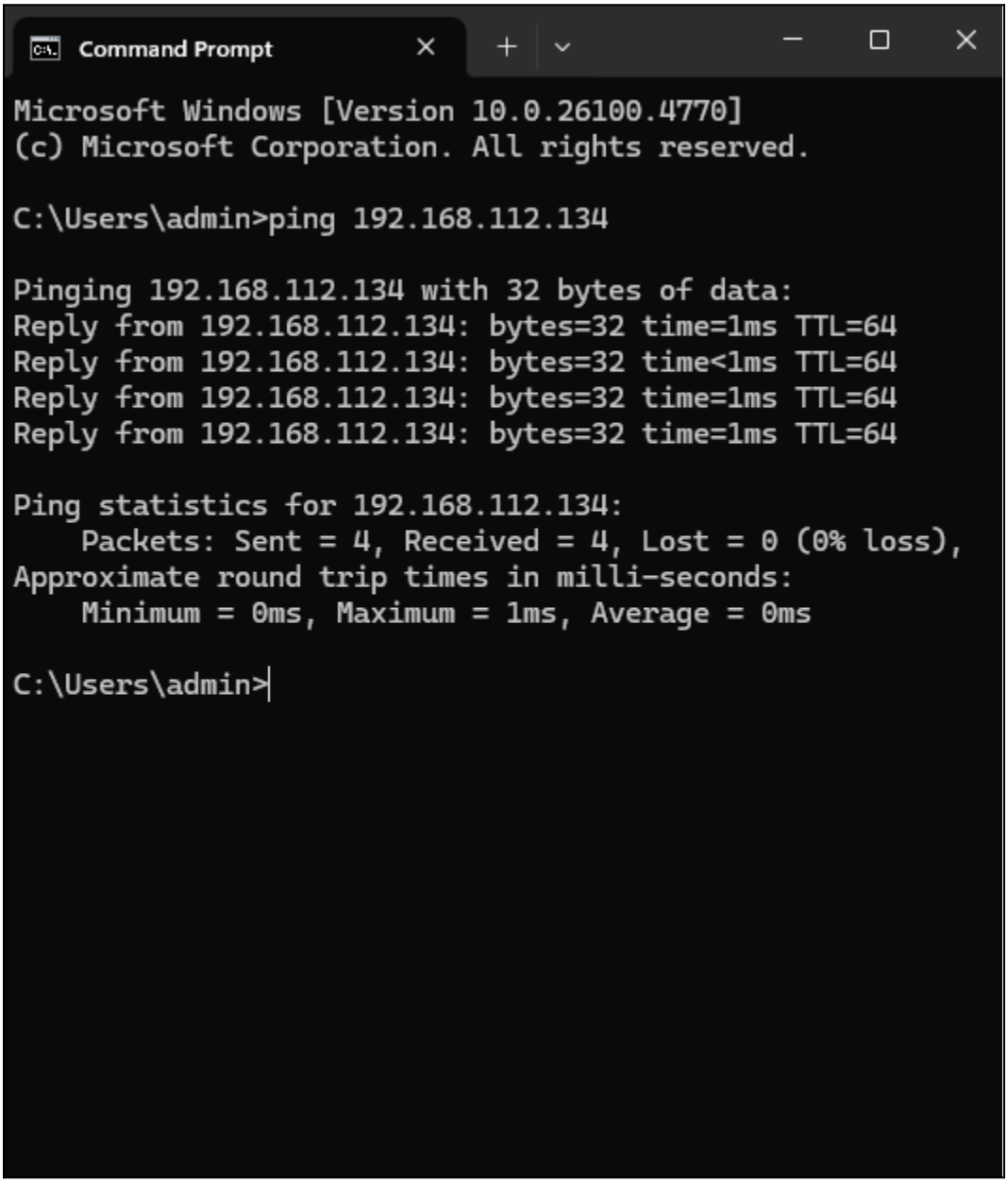
Practical 1 (03-09-25)

Working of Infrastructure as a Service.

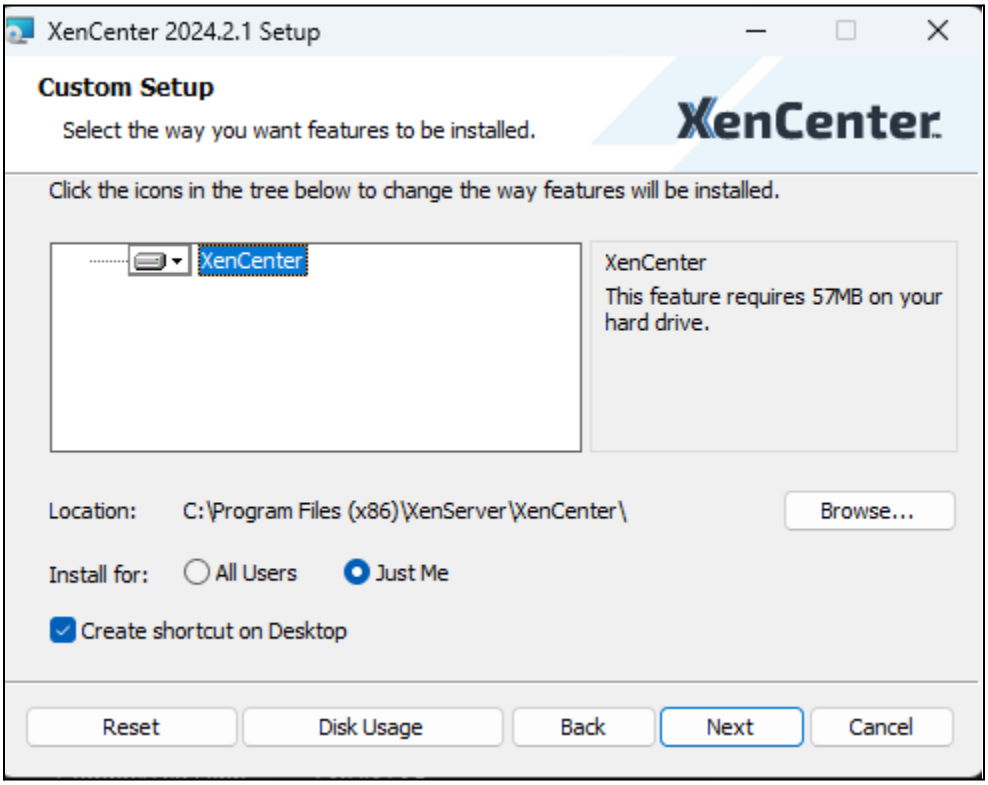
1. After creating and installing the Xen server VM, a pop-up screen will show the server's IP address,note the IP address.



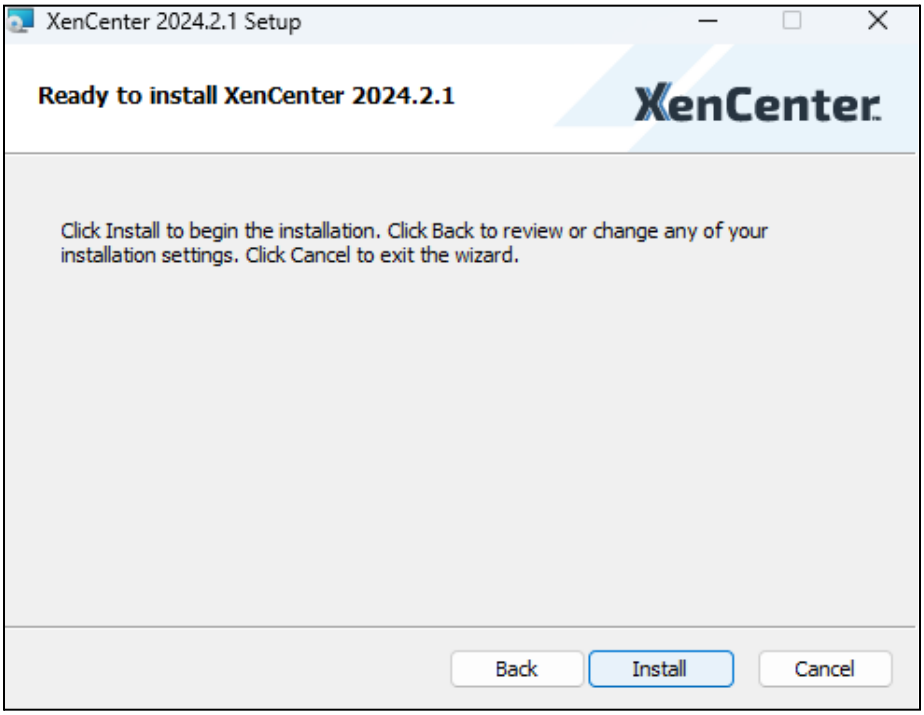
2. Go to command prompt> ping the ip address



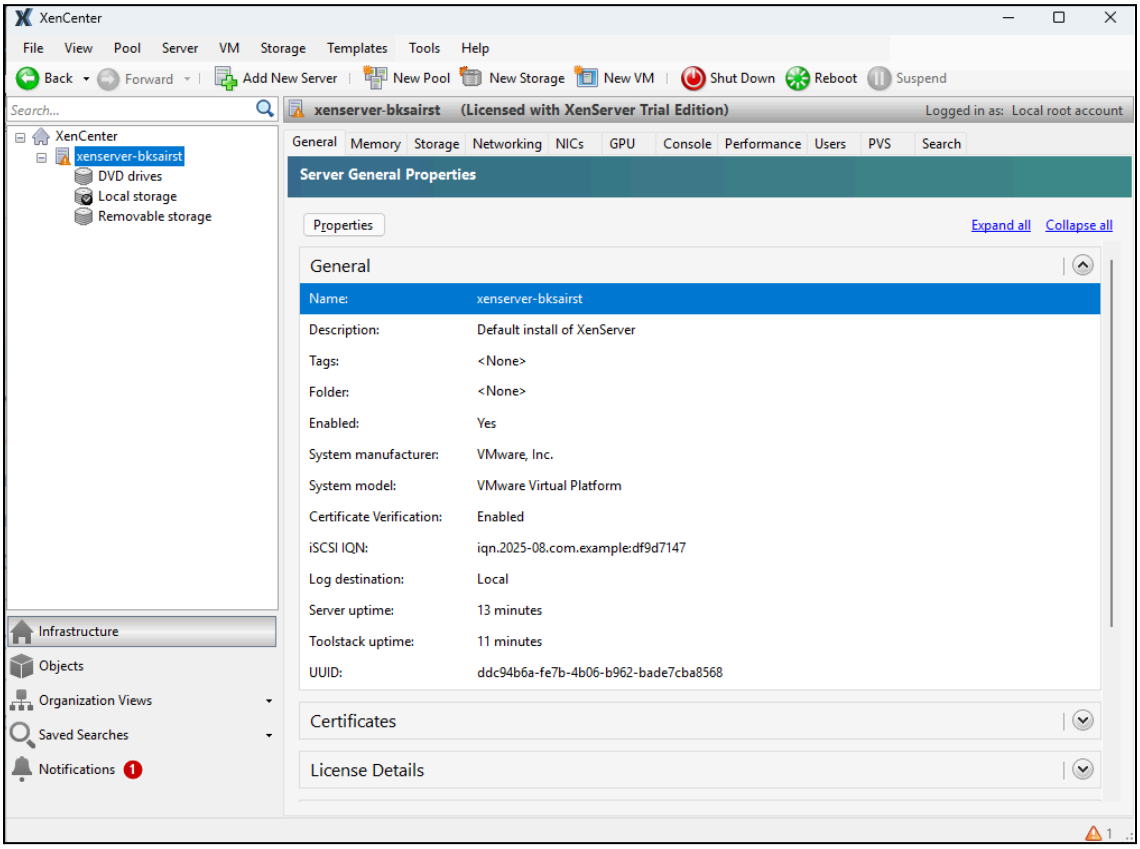
3. Download the xen Center msi file (uploaded in classroom) > keep everything as default > click next



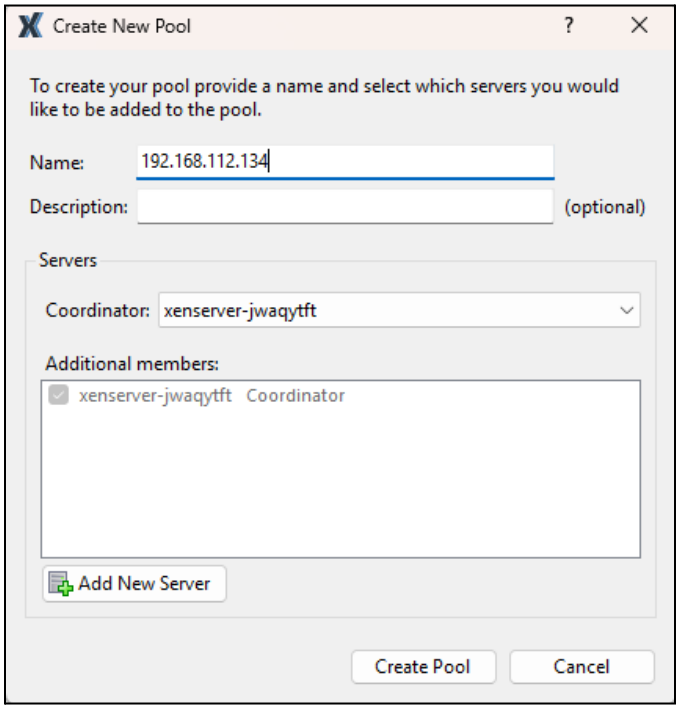
4. Click next > finish > install



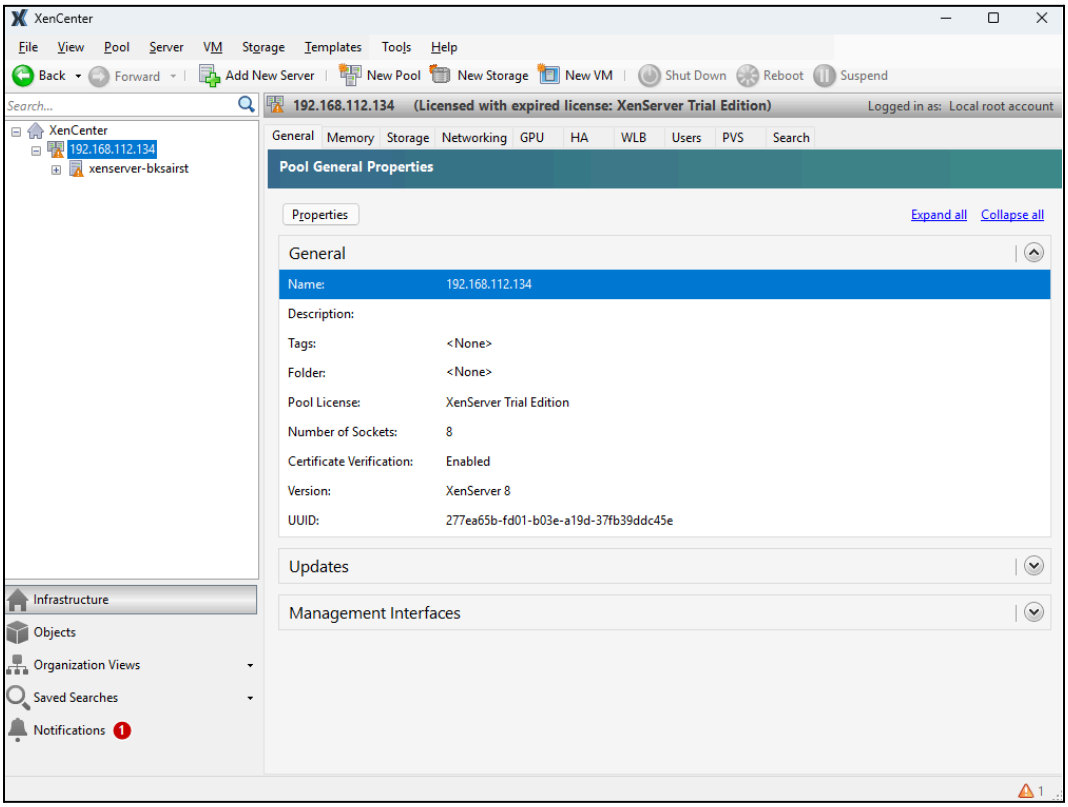
5. Click on add a server > Give your above noted ip address in server > enter the password 1234567 > add > new window will pop up keep it as default > click ok.



6. Go to new pool > create new pool > give the ip address as name > create pool.



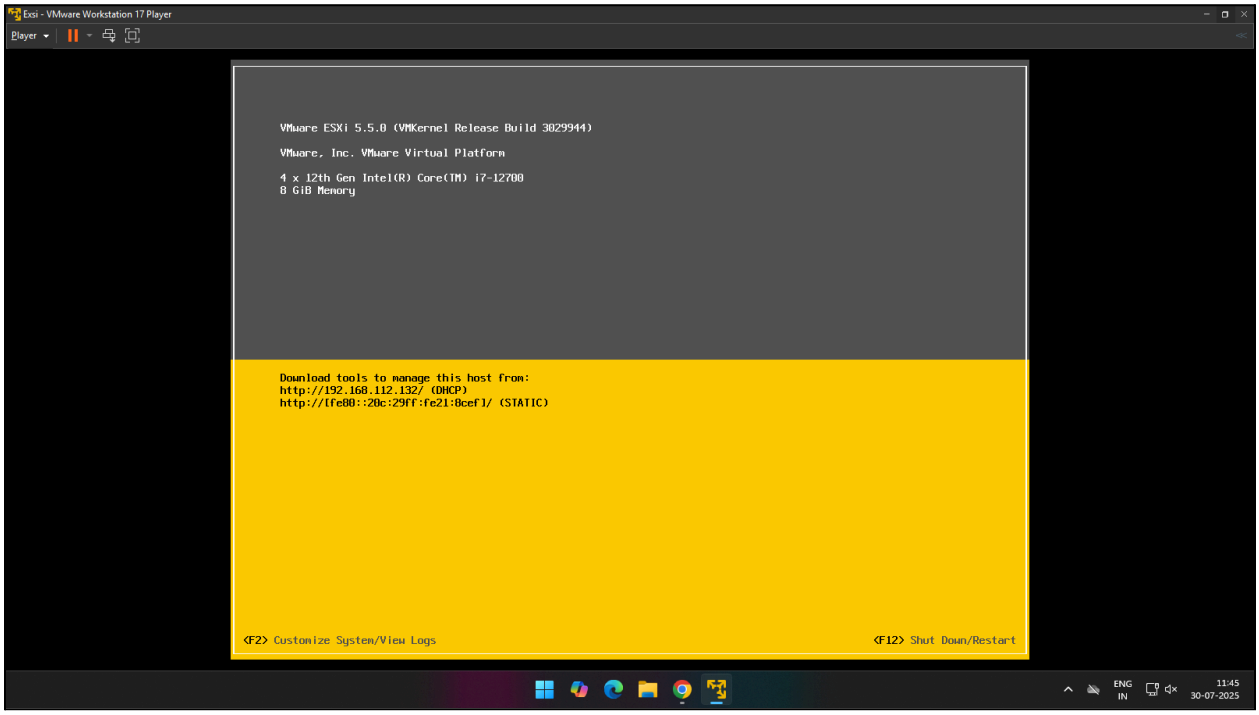
7. New pool is created

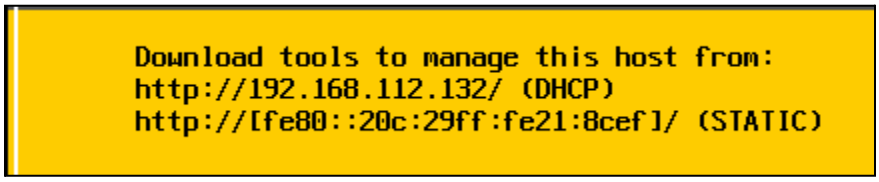


Practical 2 (10-09-25)

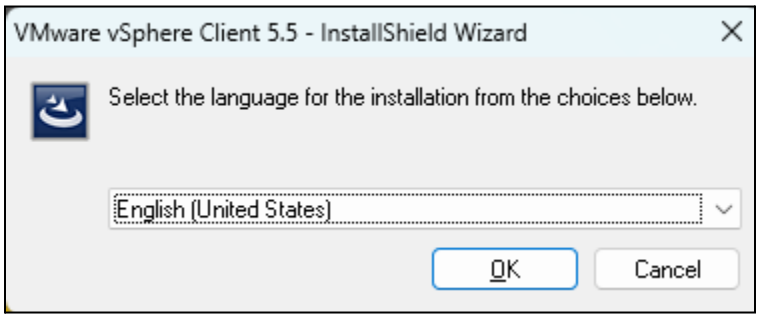
Working of Storage as a Service.

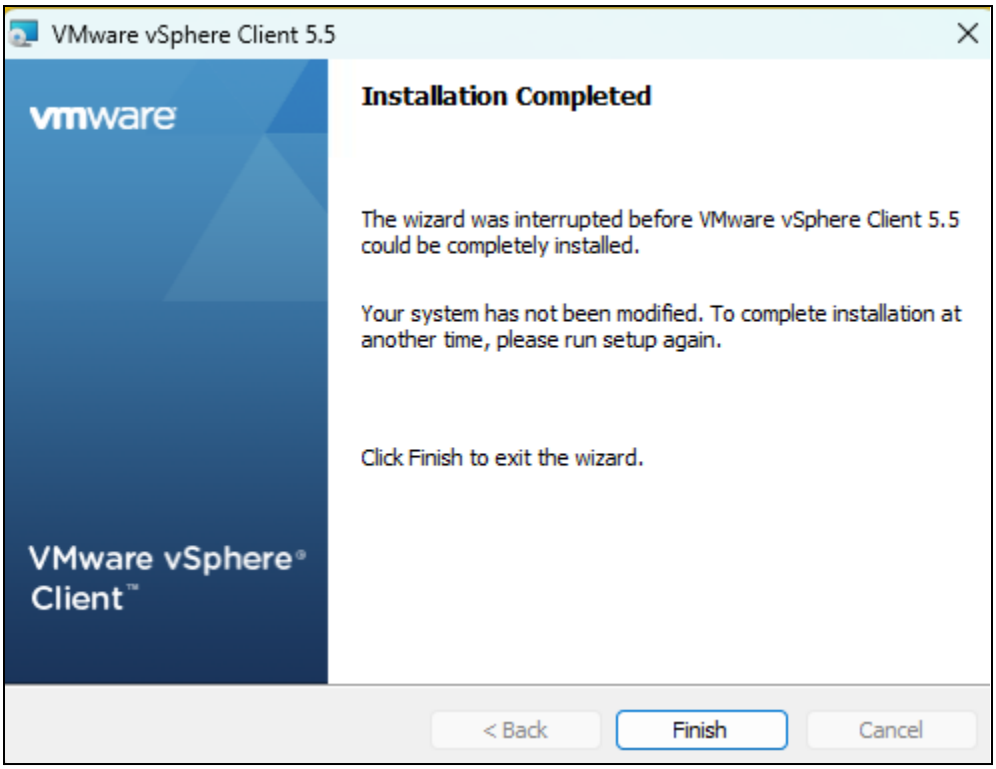
1. Create your ESXi Server > Start your VM > Note down(remember) your IP address.



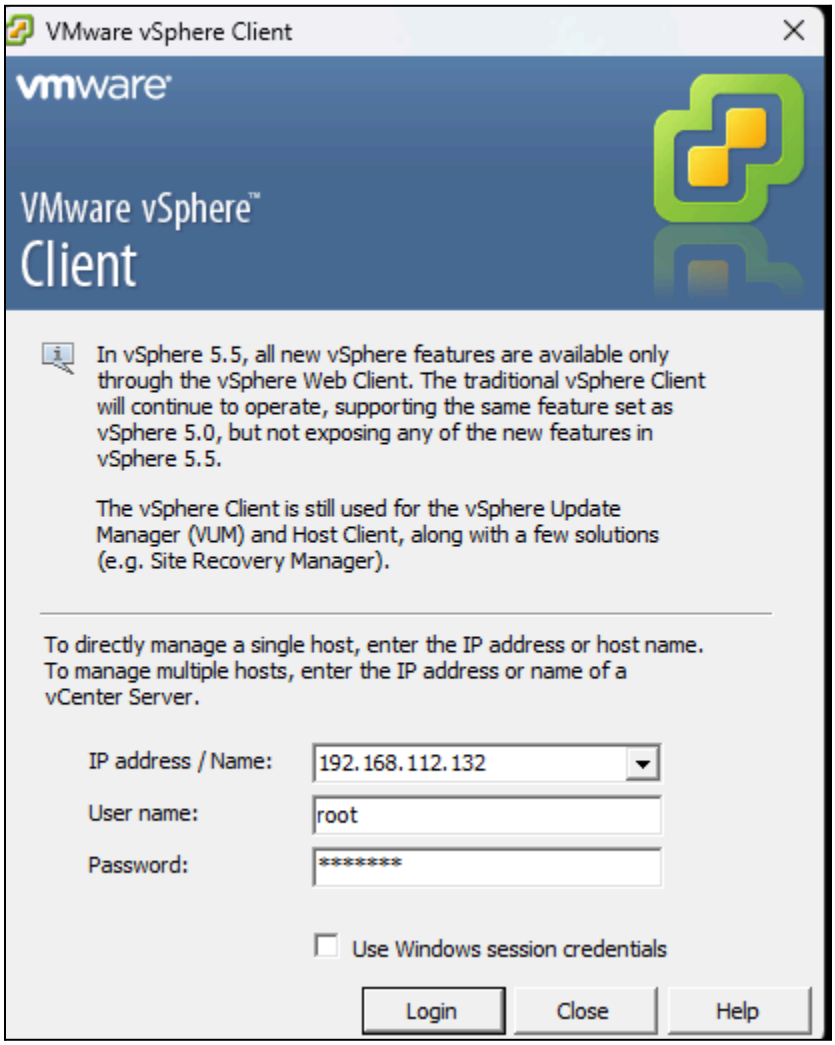


2. Install VMWare Vsphere client > Select English language > setup and install.

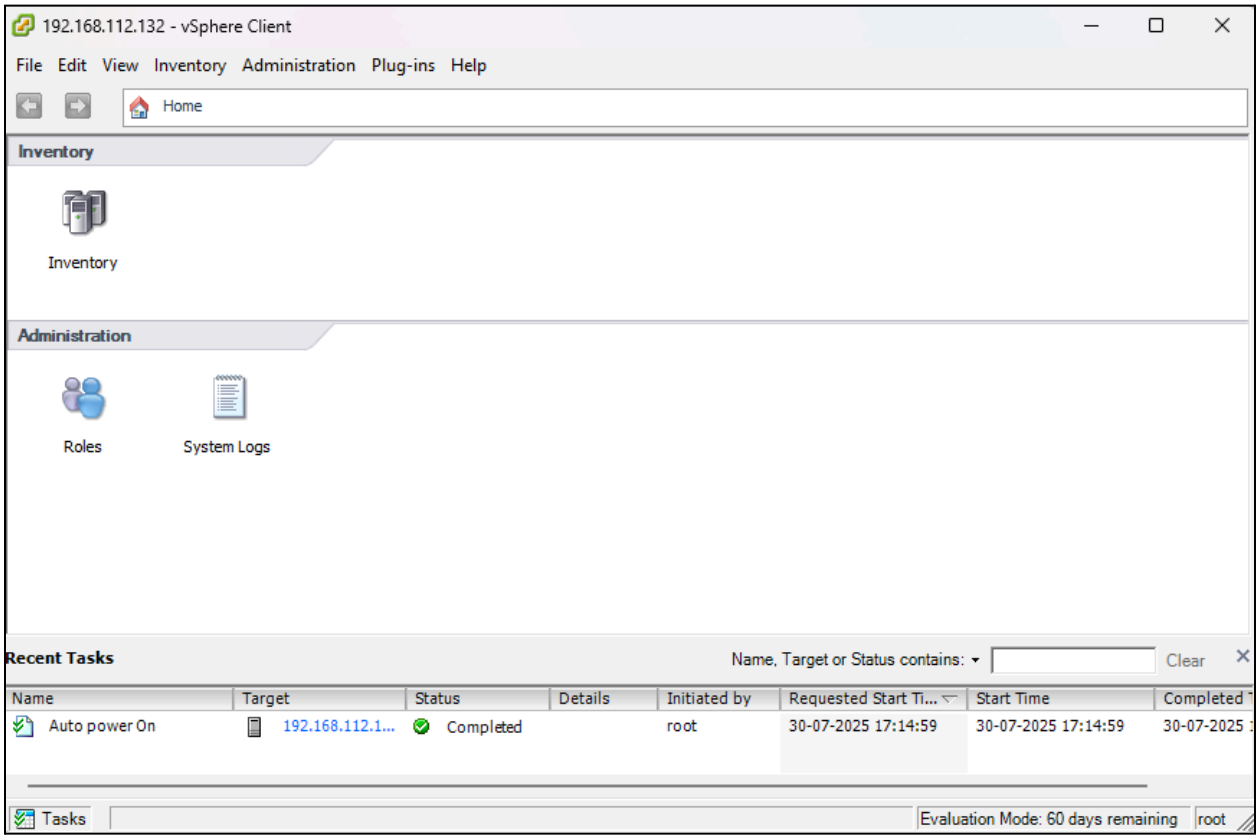




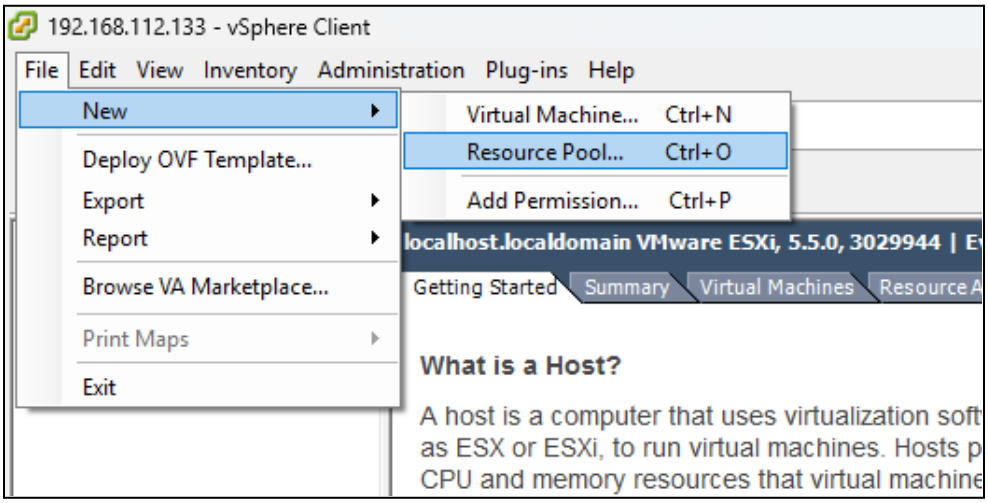
3. Give the ip address we previously noted down.
   Username: root
   Password: 1234567



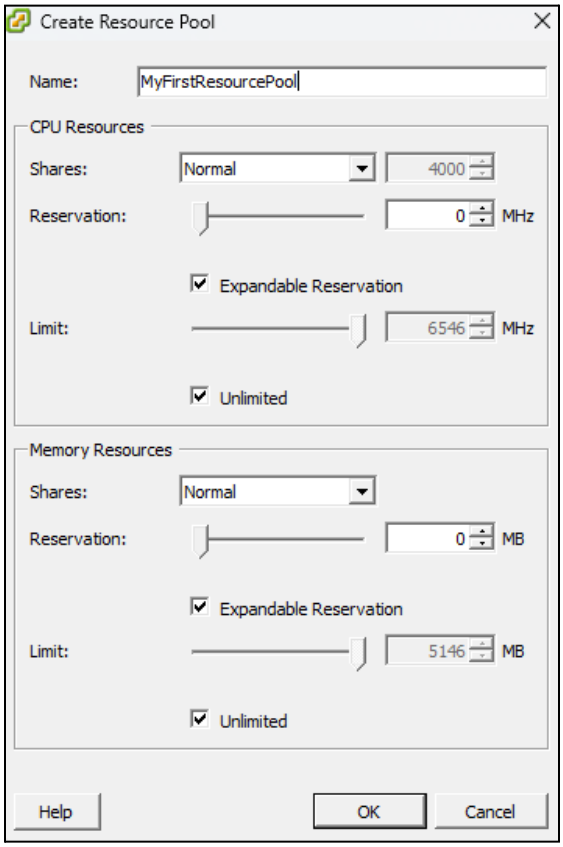
4. Click on Inventory



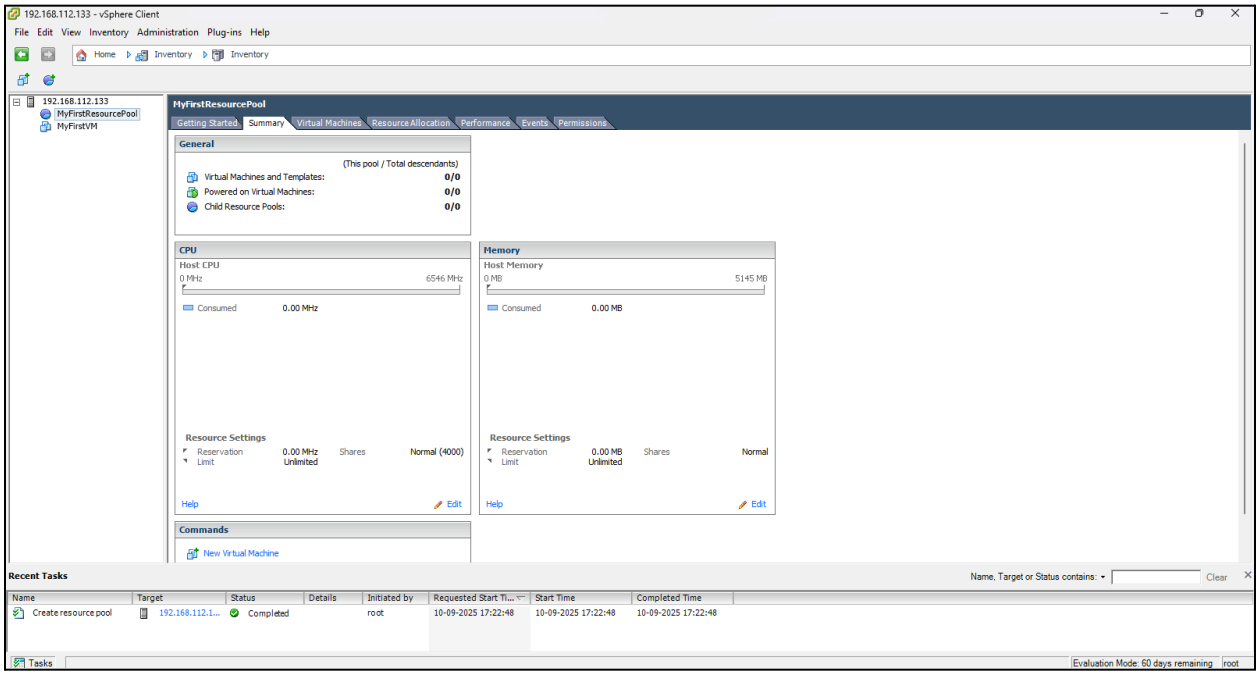
5. Click on File the New: Resource Pool



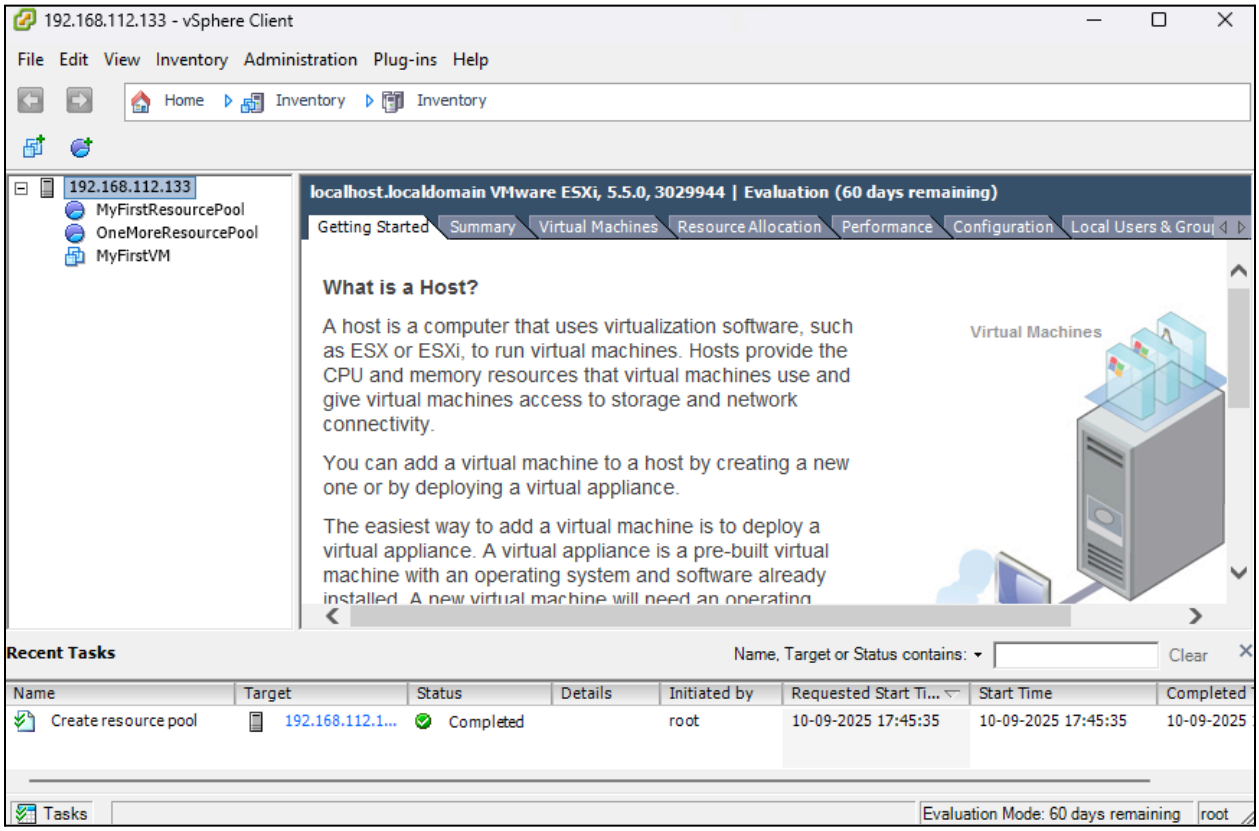
6. Create Resource Pool, keeping the default setting as it is



7. Click OK. Your pool is created!



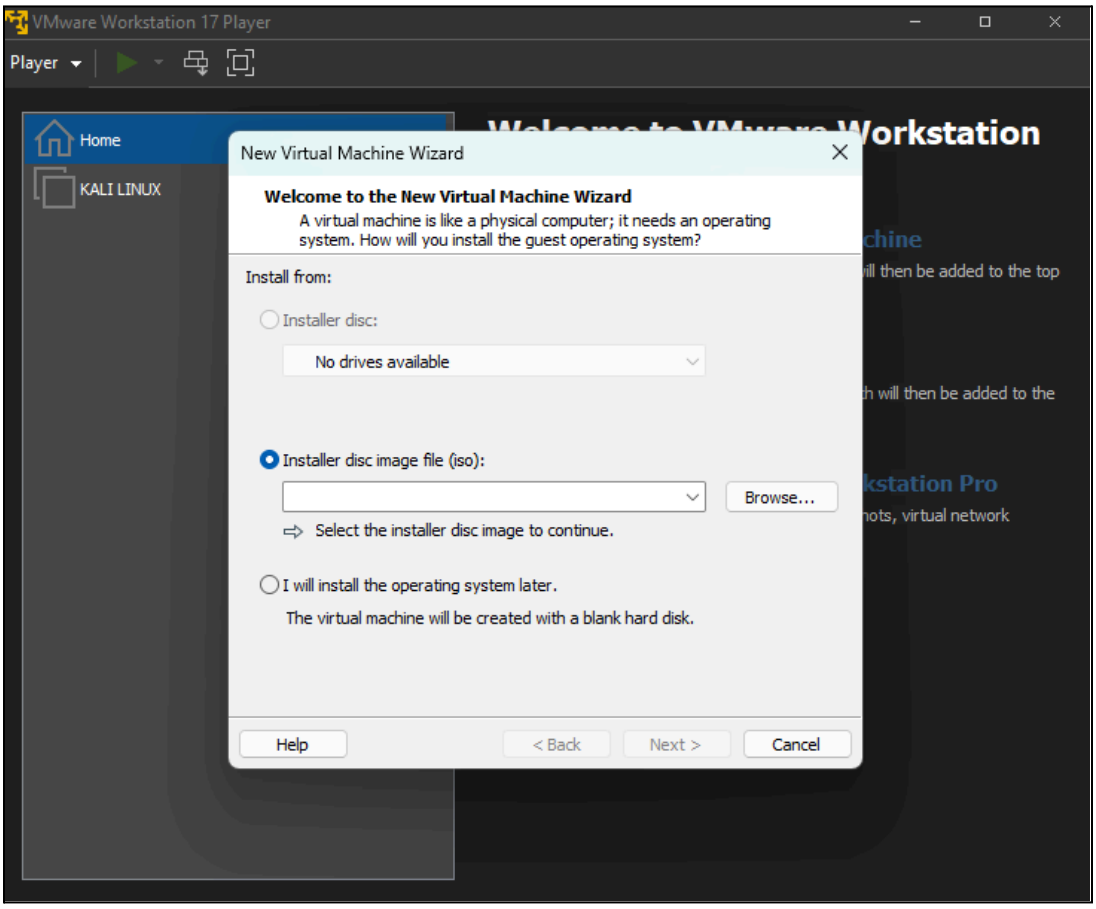
8. Similarly, we can create multiple resource pools for storage purposes



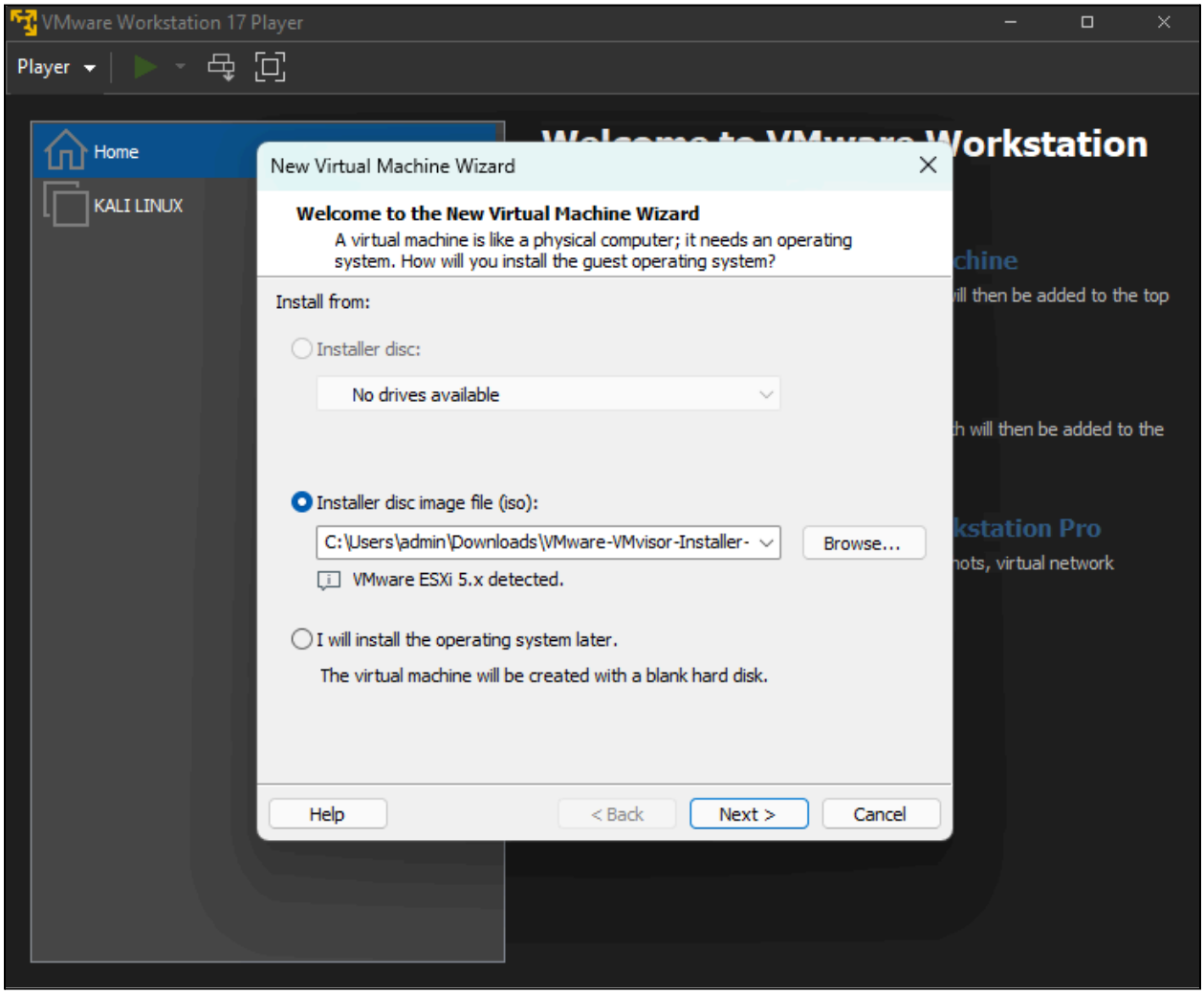
Practical 3 (16-07-25)

Implement VMWAre ESXi Server

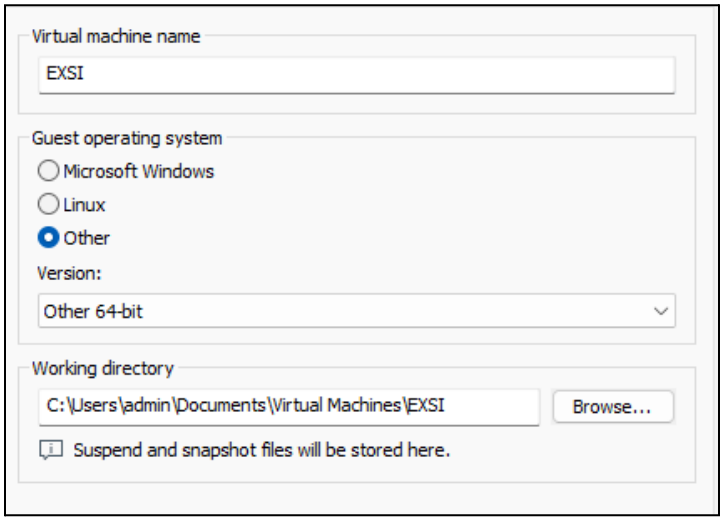
1. Go to VMWare > Create a new virtual machine.

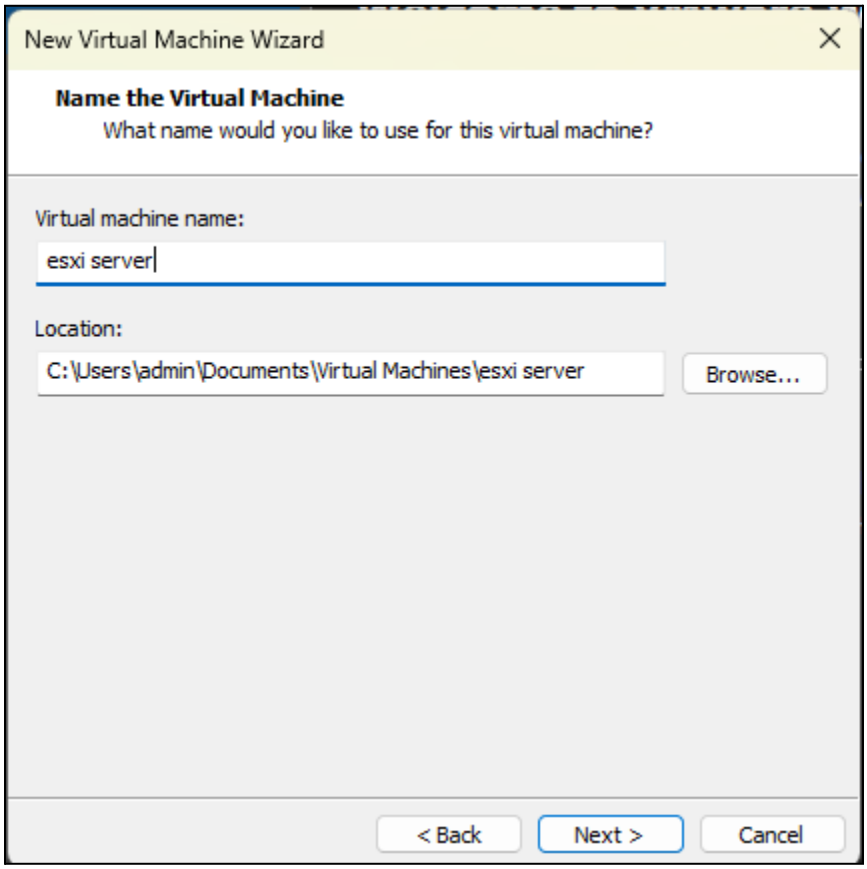


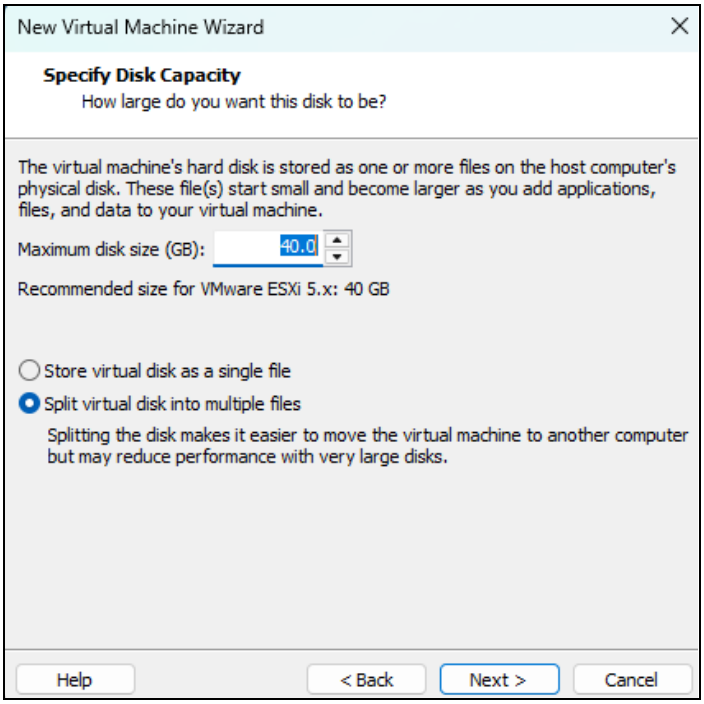
2. Browse > select the iso file (that is downloaded from classroom) > click next.

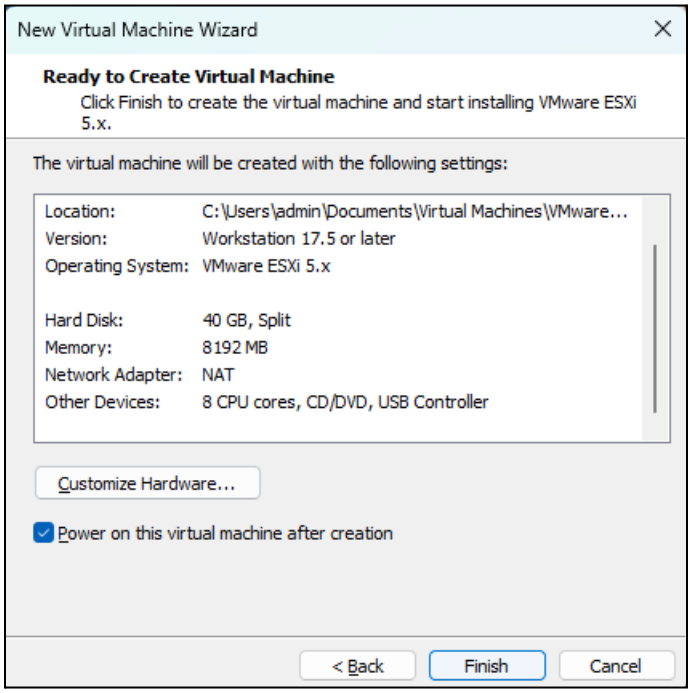


3. After clicking next > Keep everything as default (change name to exsi)



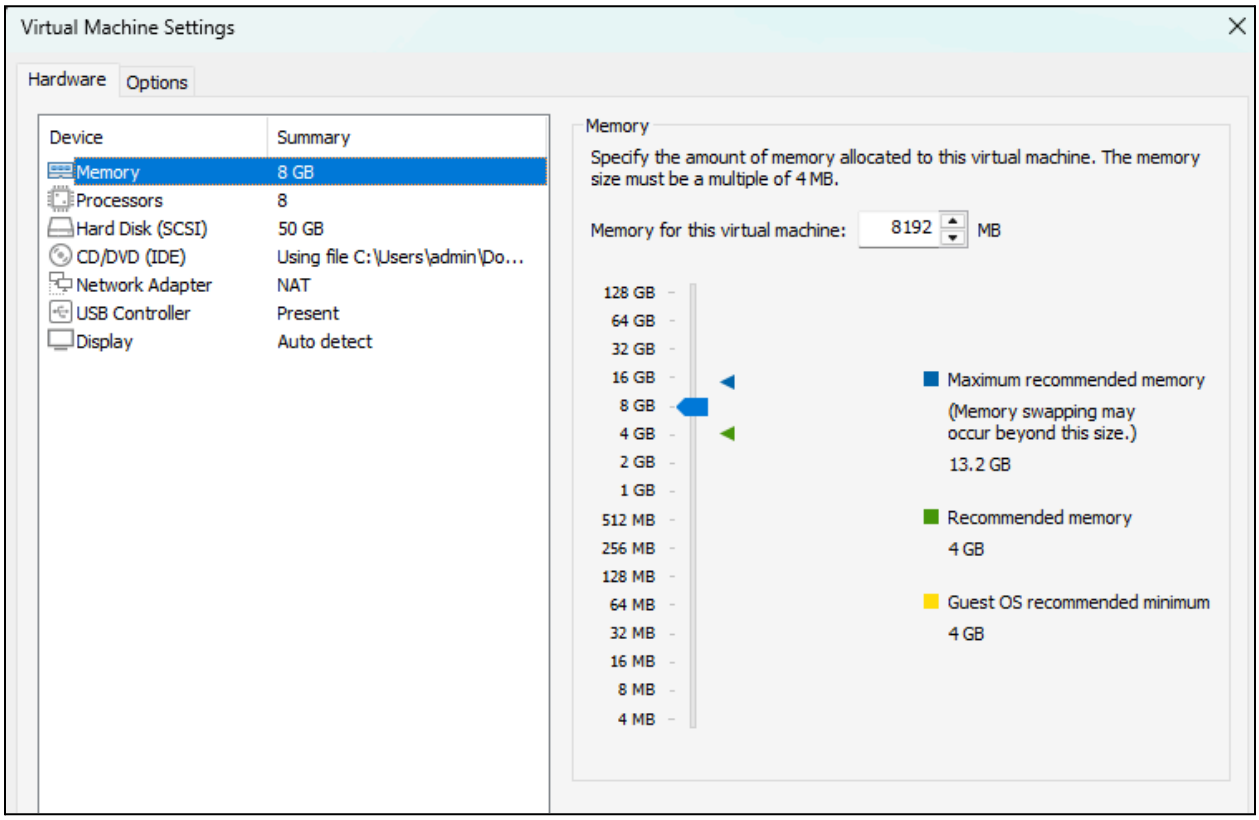



New Virtual Machine Wizard
Specify Disk Capacity
How large do you want this disk to be?
The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.
Maximum disk size (GB):
40.0
Recommended size for VMware ESXi 5.x: 40 GB
Store virtual disk as a single file
Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.
Help
< Back
Next >
Cancel

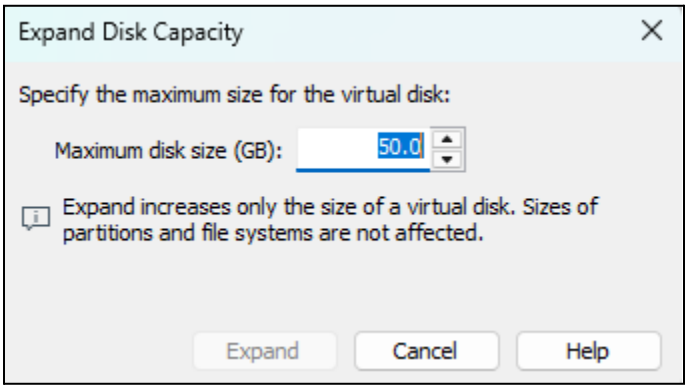

New Virtual Machine Wizard
Ready to Create Virtual Machine
Click Finish to create the virtual machine and start installing VMware ESXi 5.x.
The virtual machine will be created with the following settings:
Location: C:\Users\admin\Documents\Virtual Machines\VMware...
Version: Workstation 17.5 or later
Operating System: VMware ESXi 5.x
Hard Disk: 40 GB, Split
Memory: 8192 MB
Network Adapter: NAT
Other Devices: 8 CPU cores, CD/DVD, USB Controller
Customize Hardware...
Power on this virtual machine after creation
< Back
Finish
Cancel

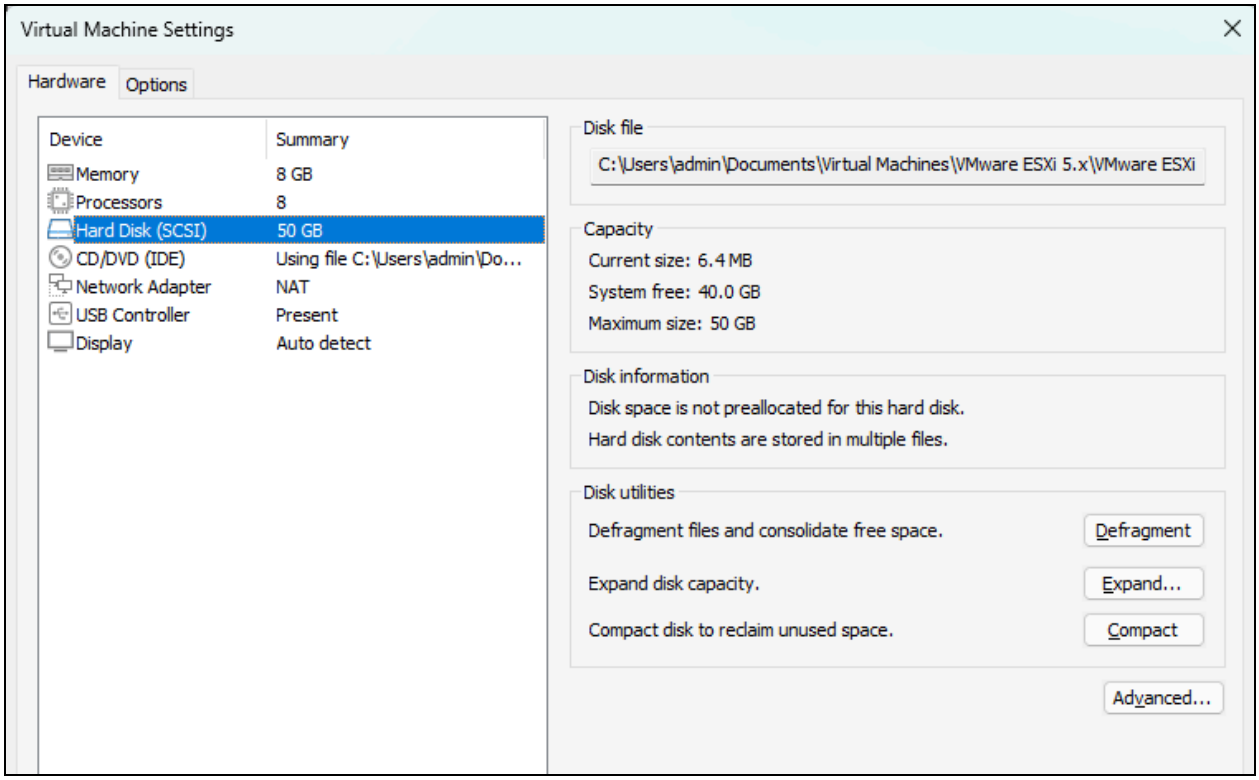
finish

4. Customize hardware > Go to memory > change it from 4GB to 8GB

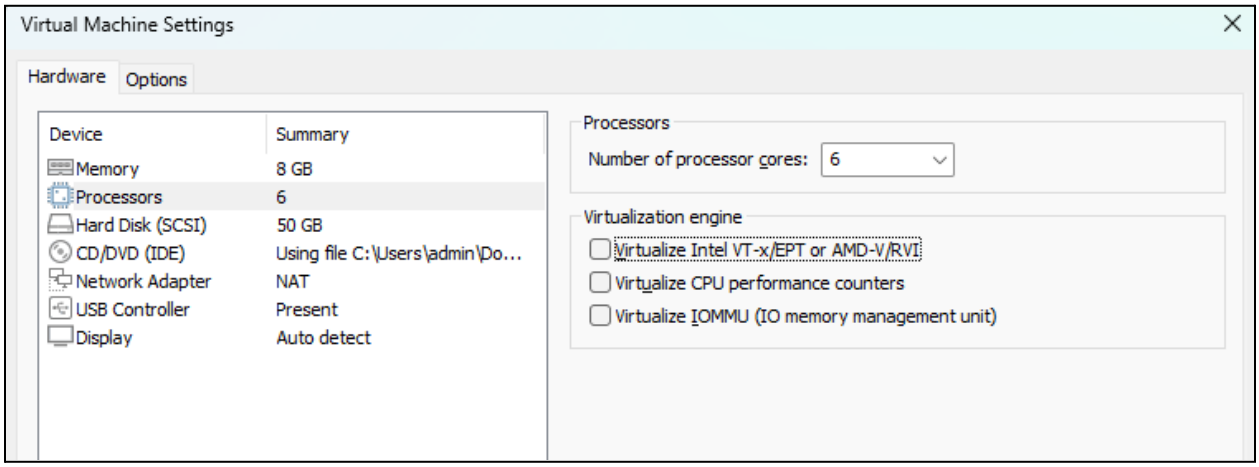


5. Go to the hard disk > expand> change it from 40gb to 50 gb.

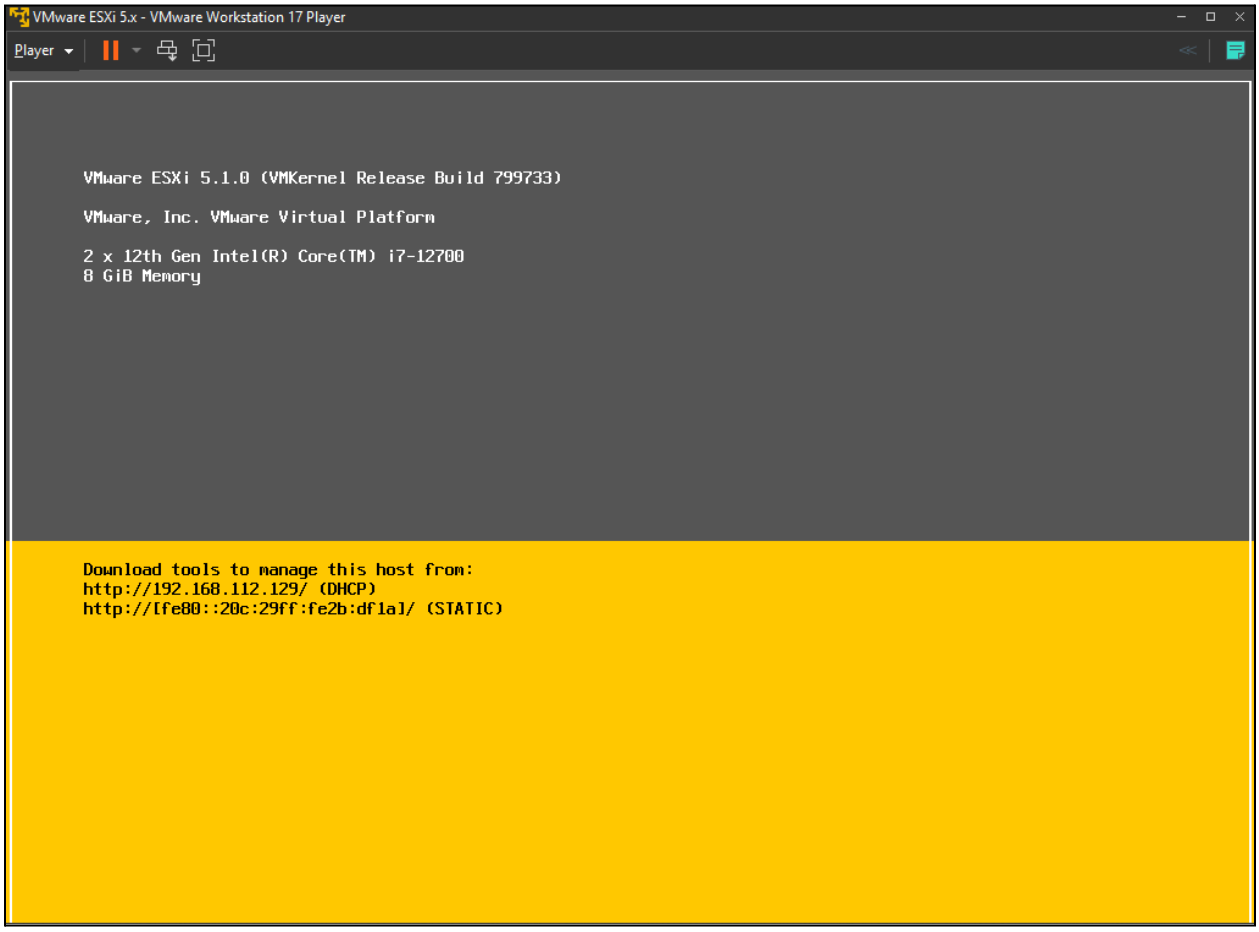




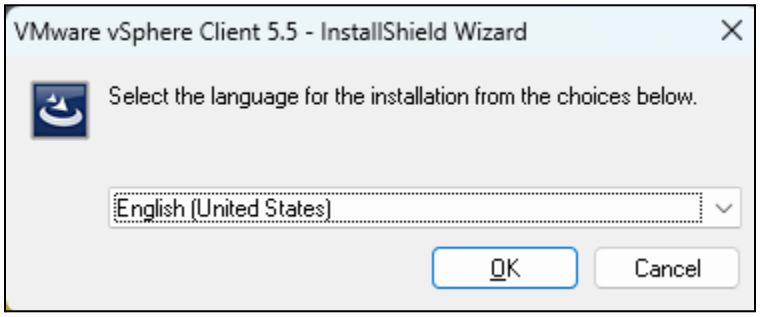
6. go to processors > Change it from 2cores to 6 cores > uncheck the virtualize intel vt-x/ept..... > click ok.

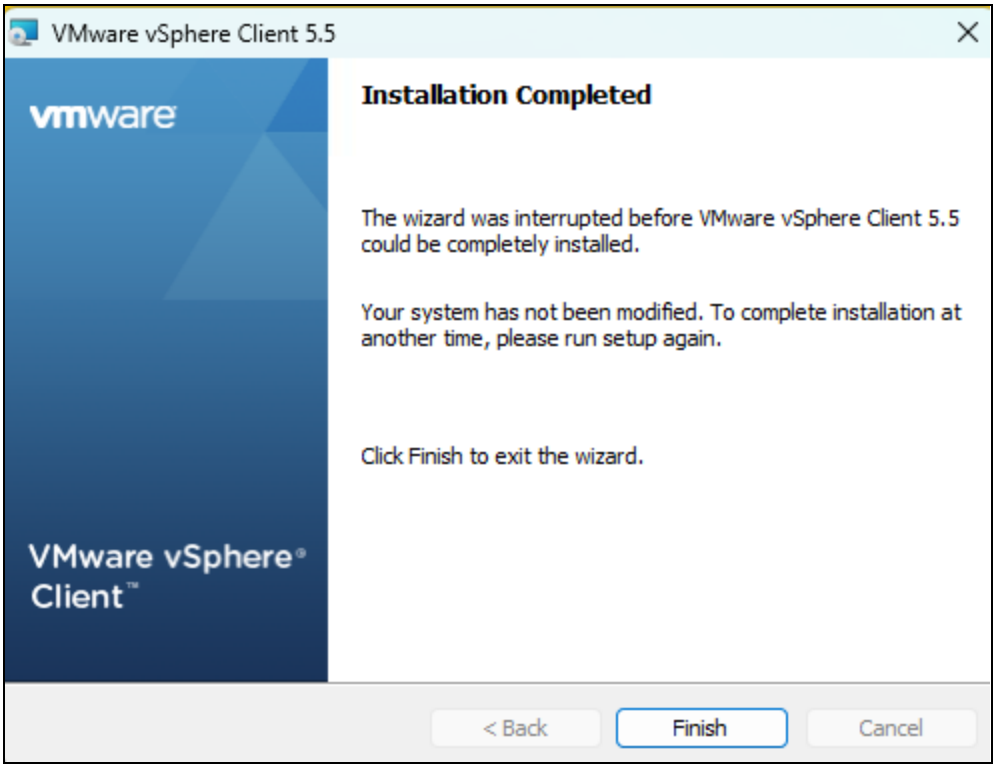


7. Start your VM > Build your VM> set password : 1234567 > Enter>Install>reboot> Note down(remember) your IP address.

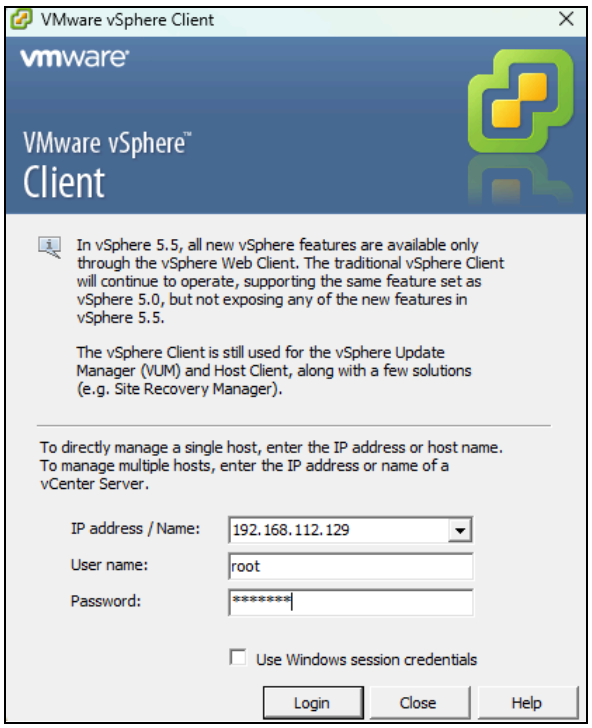


8. Install VMWare Vsphere client > Select English language > setup and install.

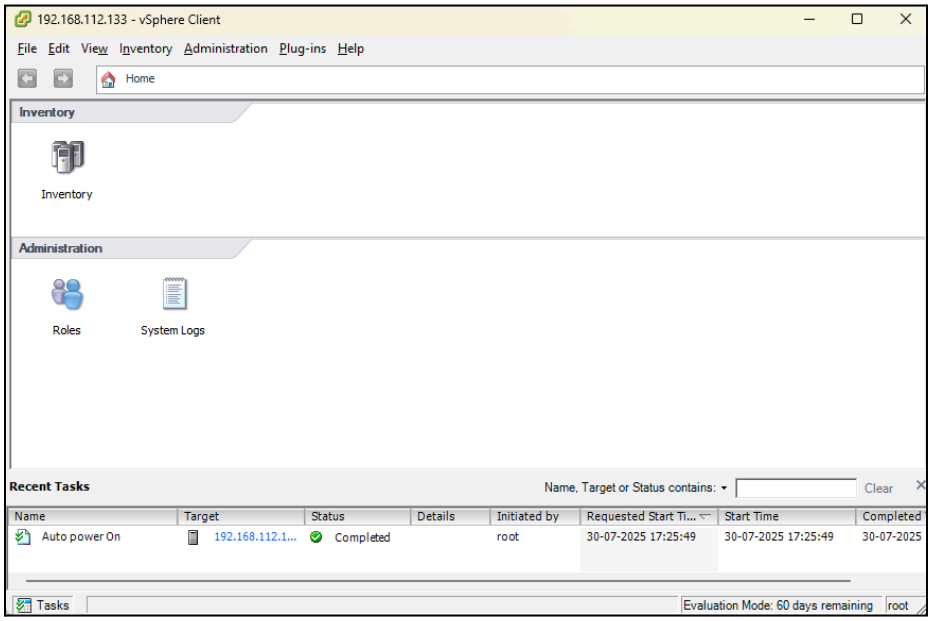




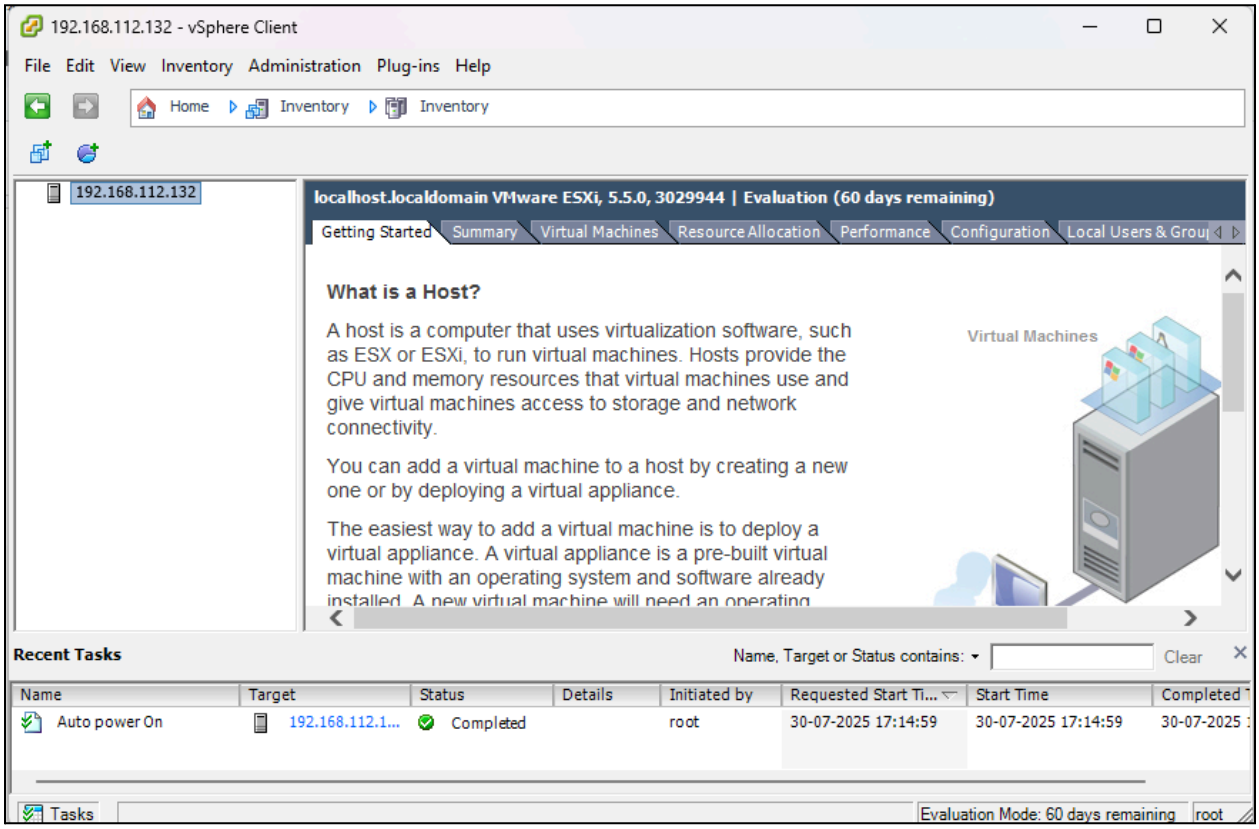
9. Give the ip address we previously noted down.
   Username: root
   Password: 1234567



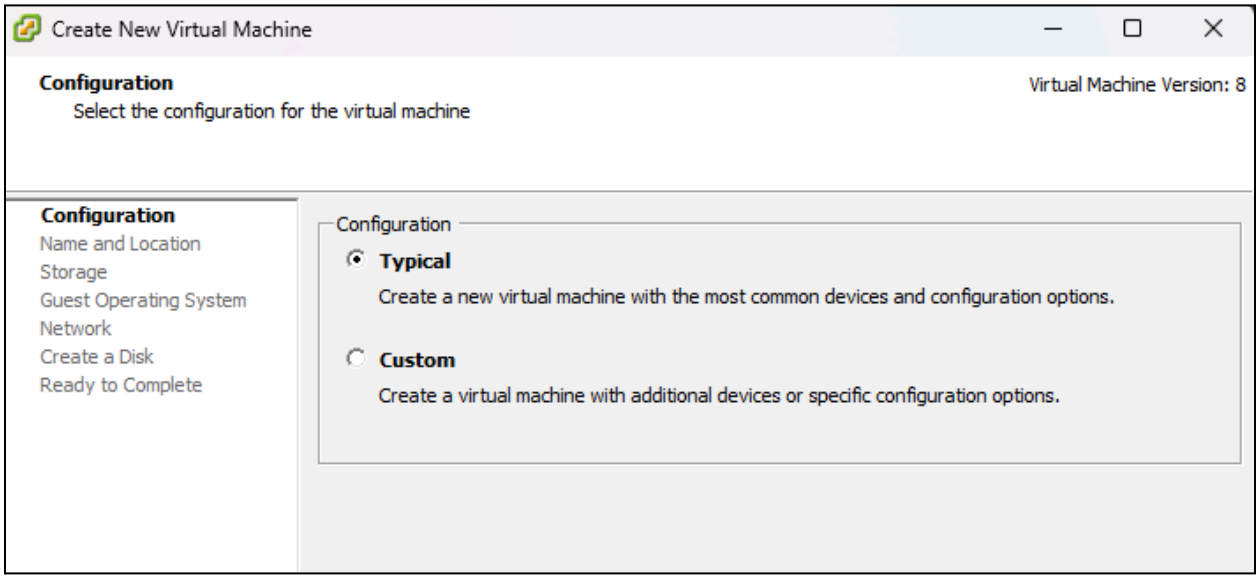
10. Click on inventory



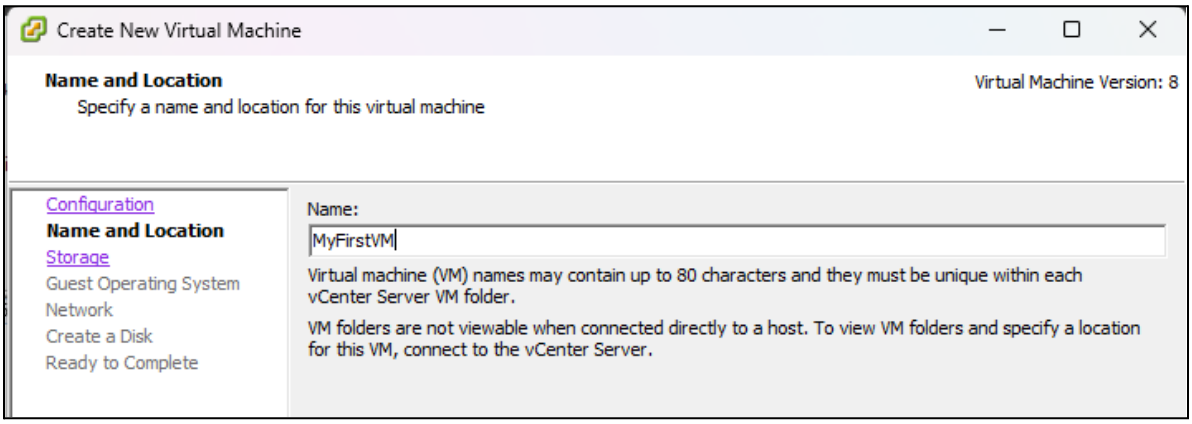
11. Go to File > New > Create New Virtual Machine



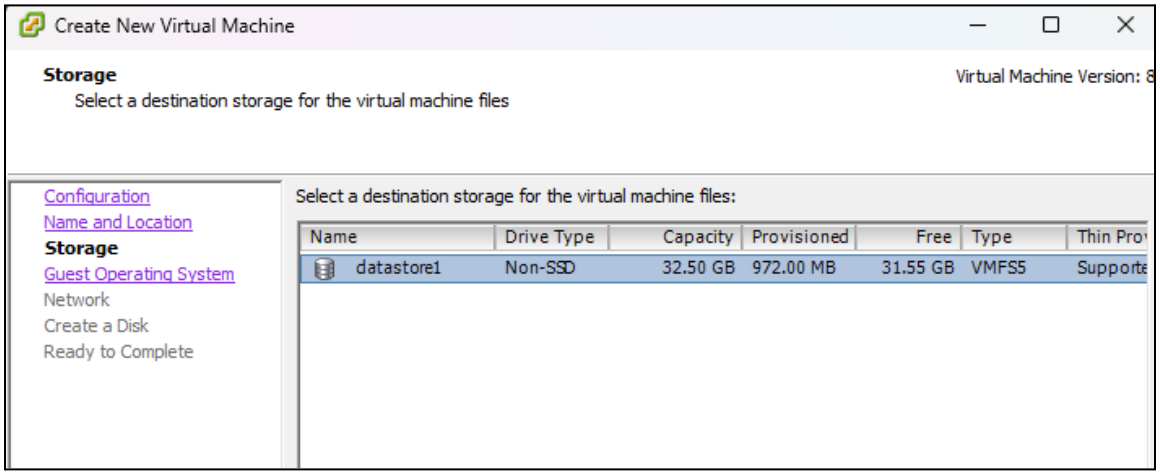
12. Select Typical > next



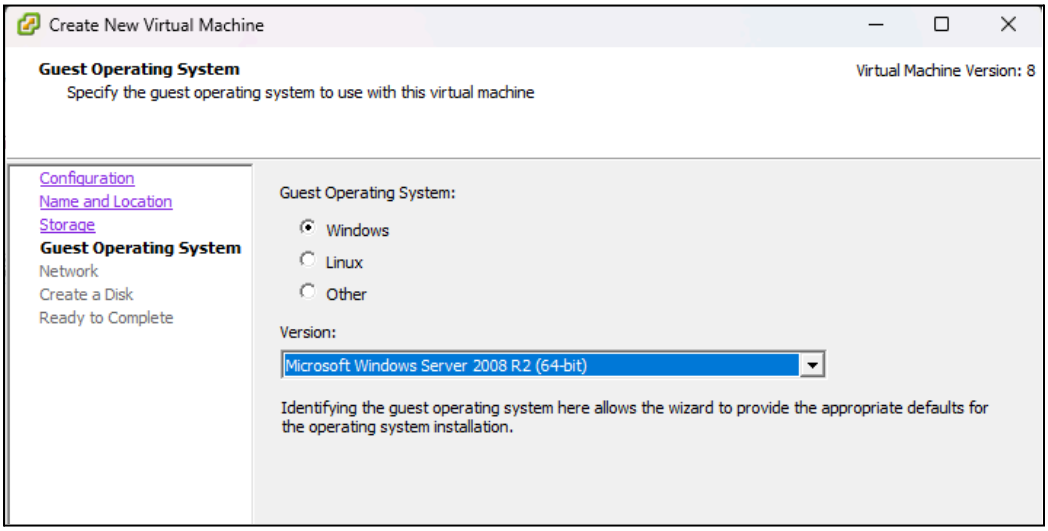
13. Click Next > Name the file



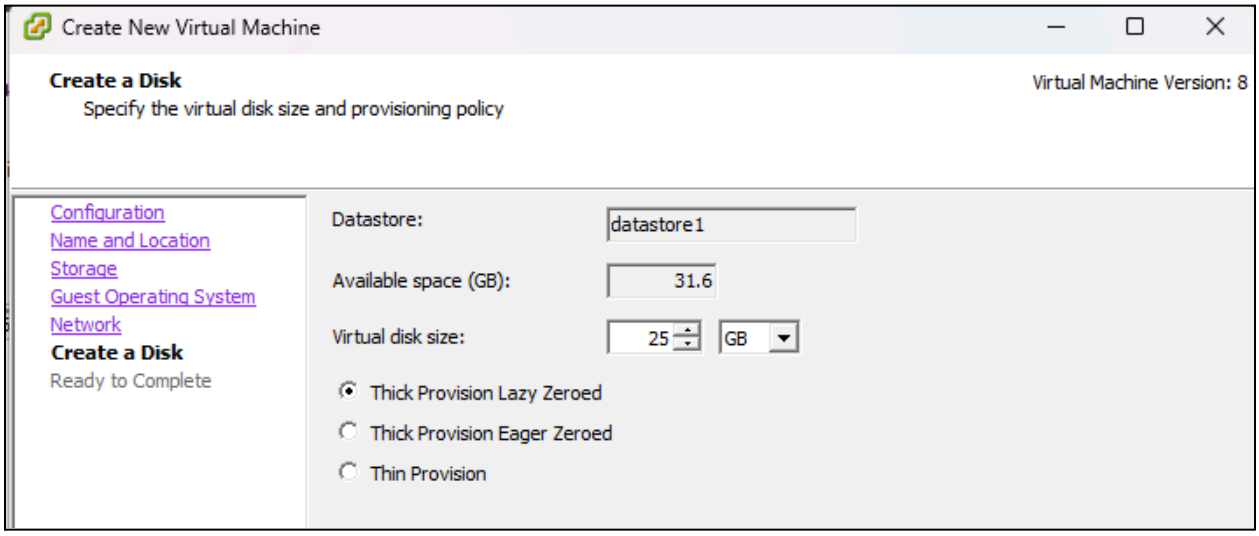
14. Click Next



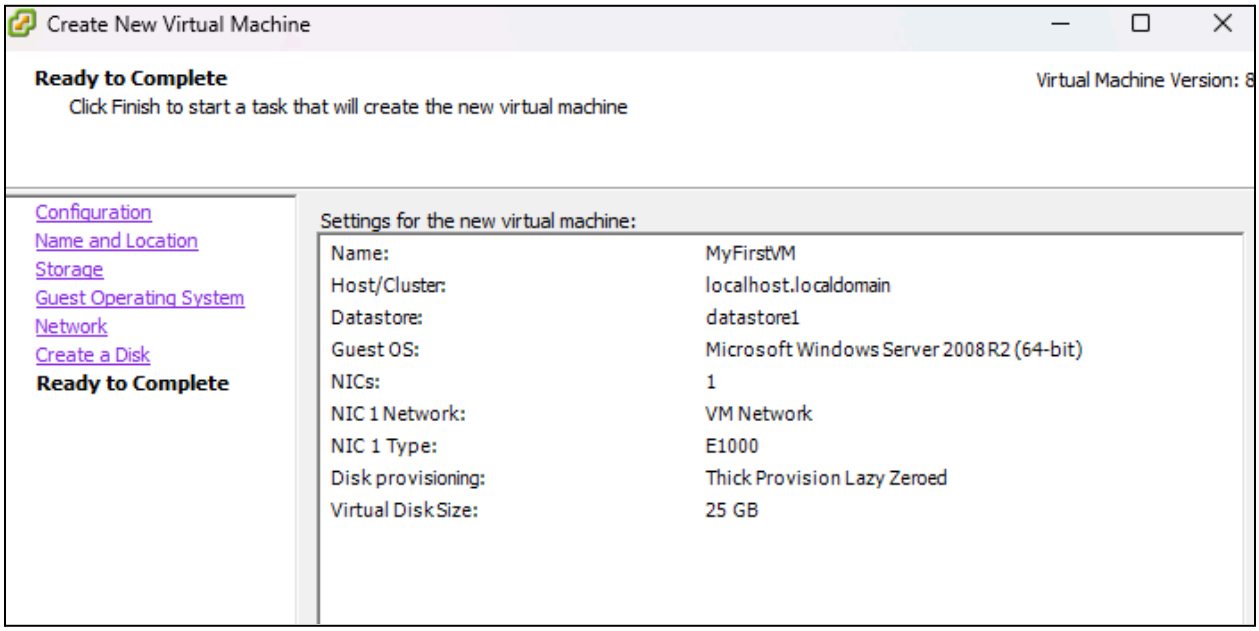
15. Click Next



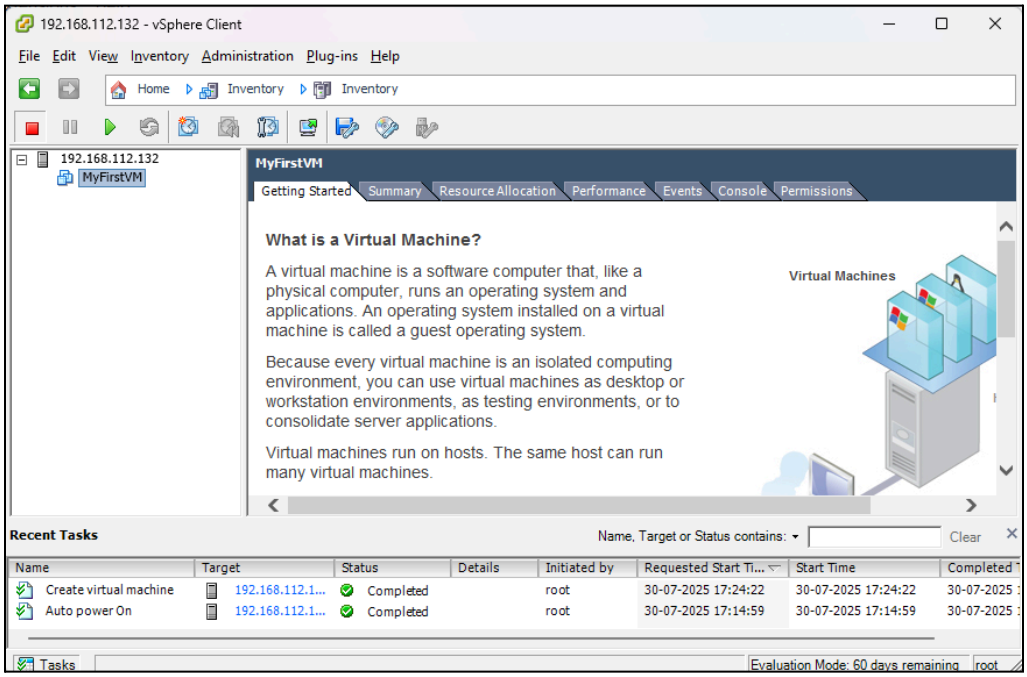
16. Click Next > Set the virtual disk size as per space available



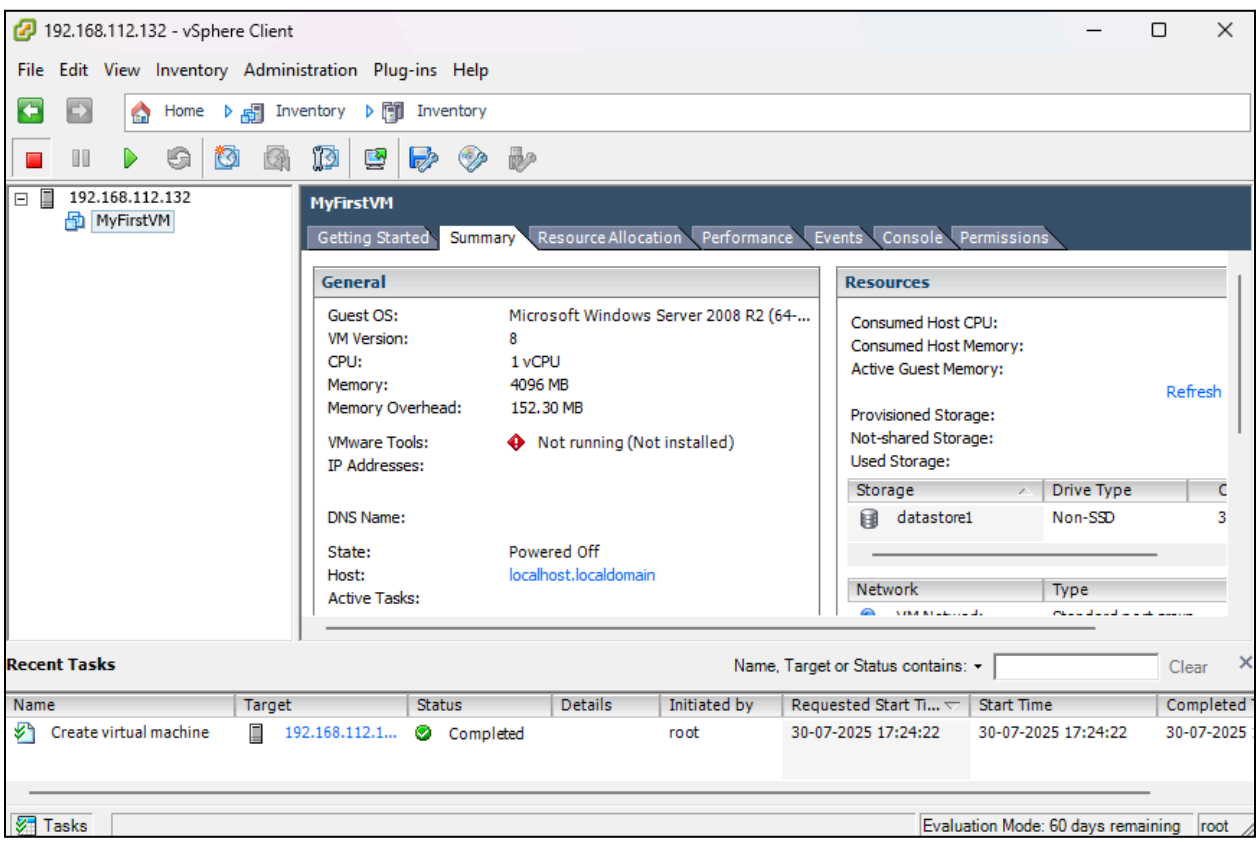
17. Click Next



18. Click Finish > Your New Virtual machine is ready!



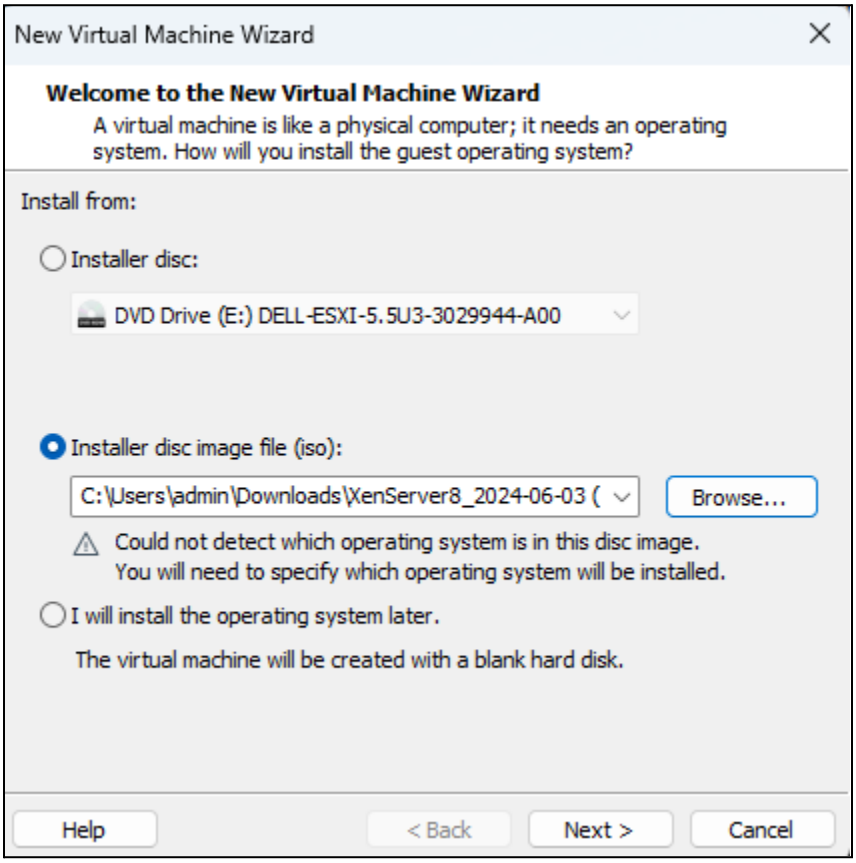
19. Click on summary to see the summary of the virtual machine created.



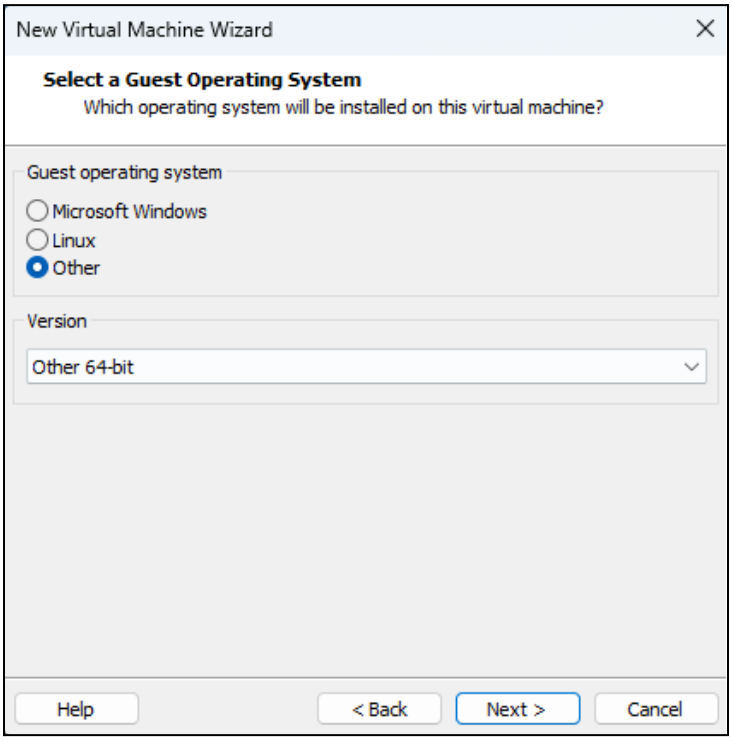
Practical 4 (06-08-25)

Managing and working of cloud Xen server.

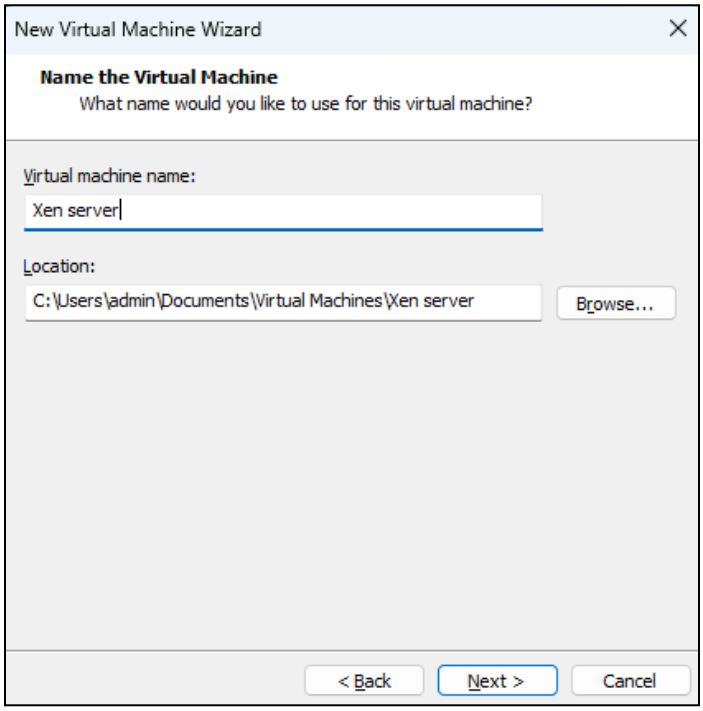
1. Go to VmWare > Create a new virtual machine > select the xen server iso file > next.



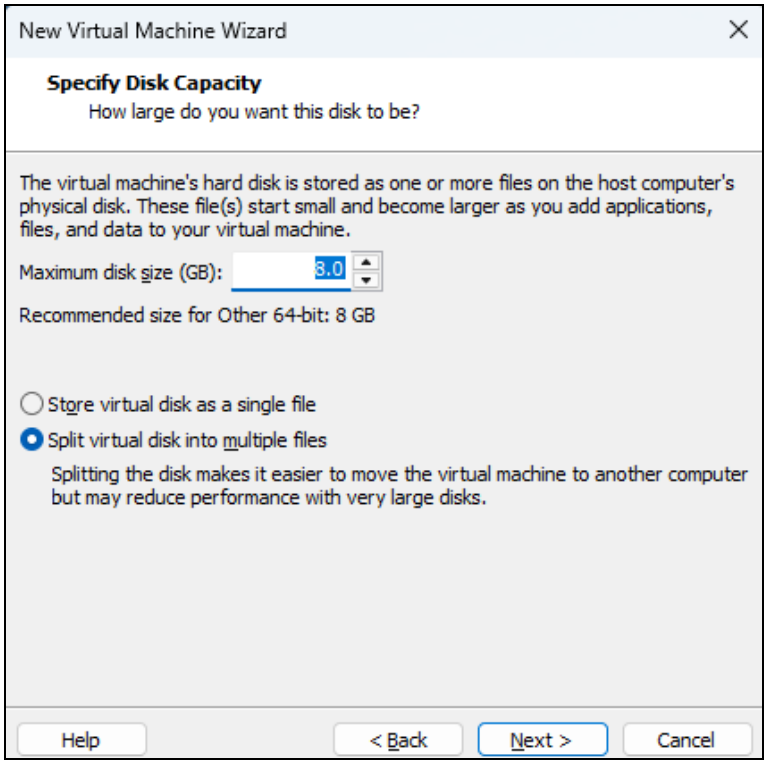
2. Select other as guest OS > Select version as other 64-bit > next



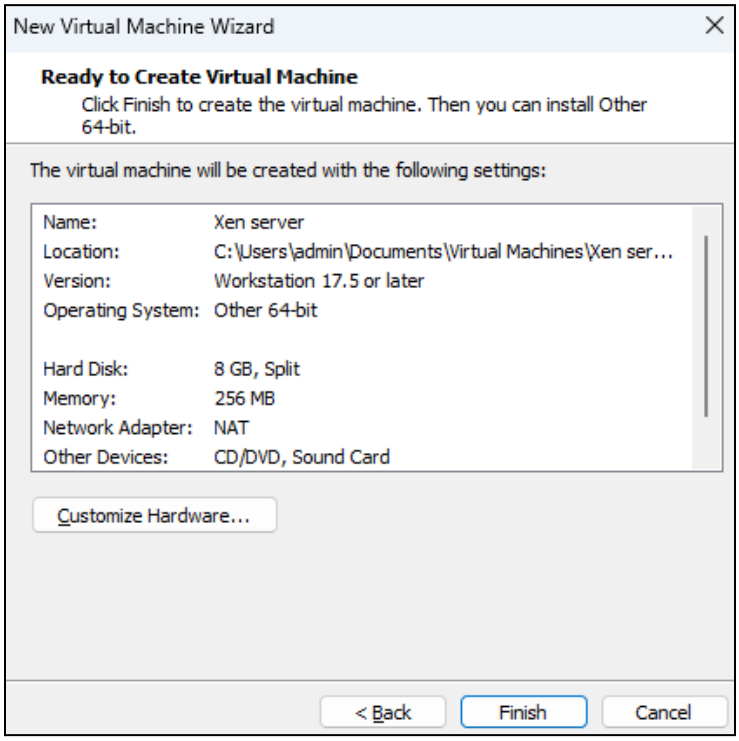
3. Name the virtual machine > next



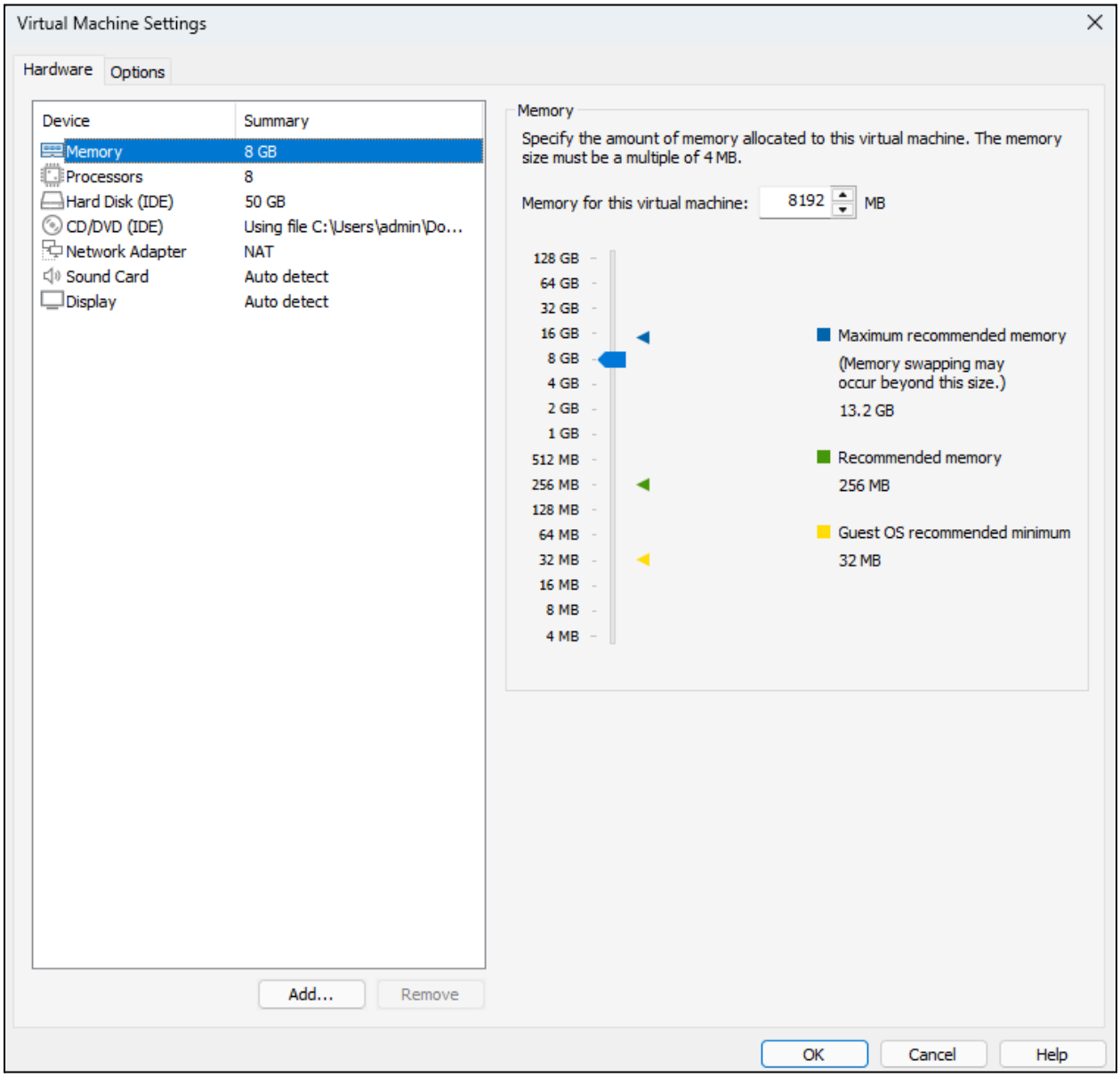
4. Click next



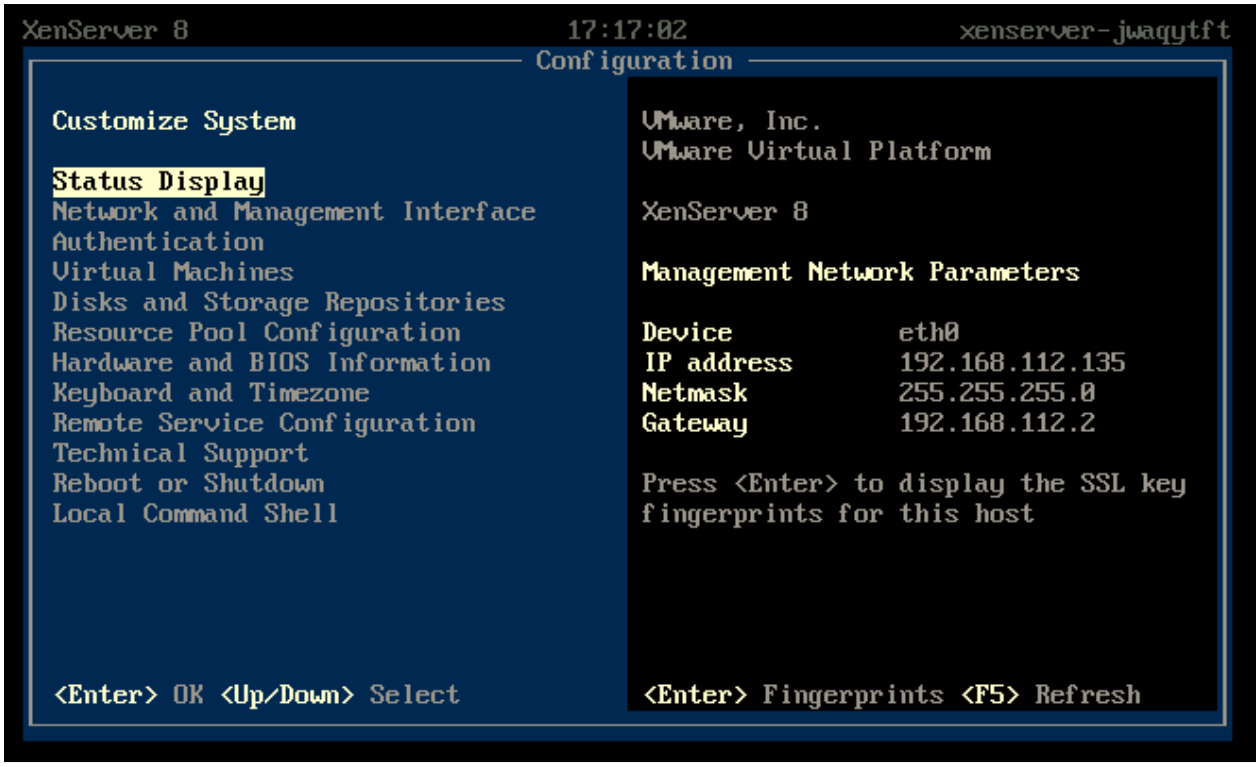
5. Click next > finish



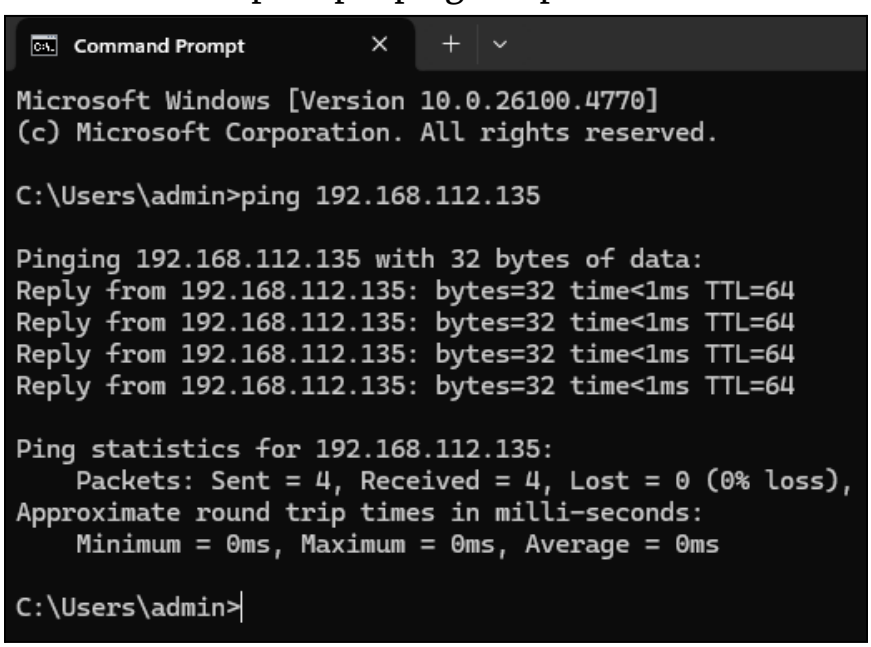
6. Go to Edit Virtual machine > Set the memory to 8 GB > set processors to 8 > Expand the disk size to 50 GB > Click OK > Play the VM



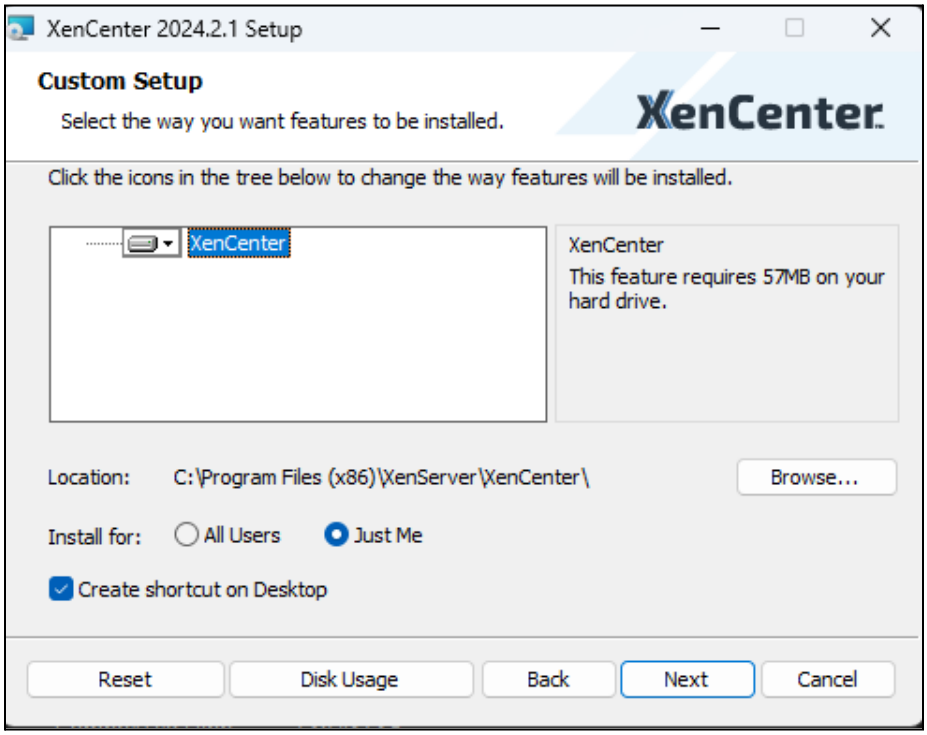
7. Play the VM > set up > Keep Clicking ok > set password to 1234567 > Set time zone to Asia > set city to kolkata > select DHCP server > install xen server > After installing > Click NO on installing any supplementary pack > reboot > after rebooting this screen will pop-up > note the IP address



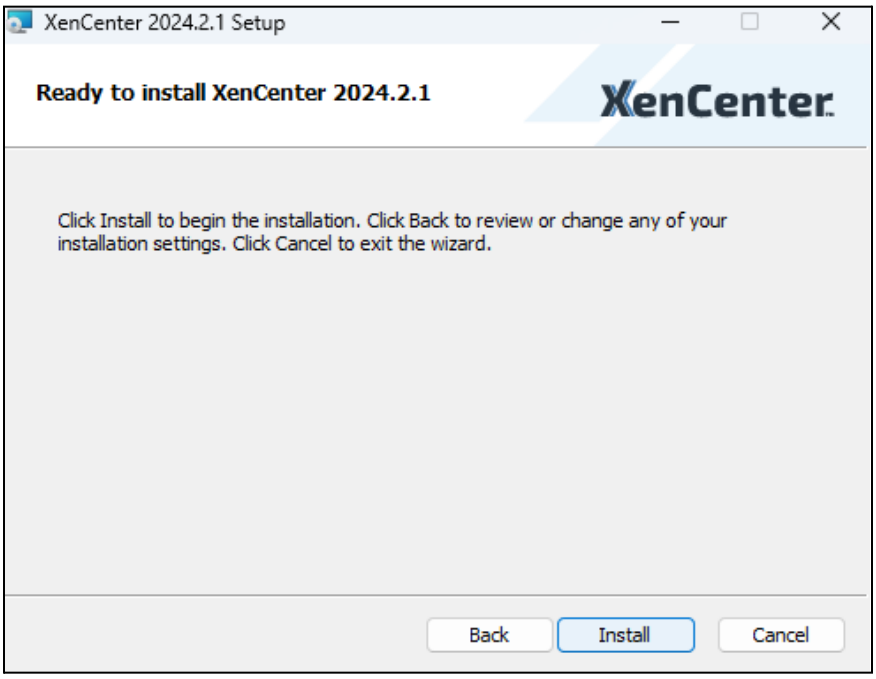
8. Go to command prompt> ping the ip address



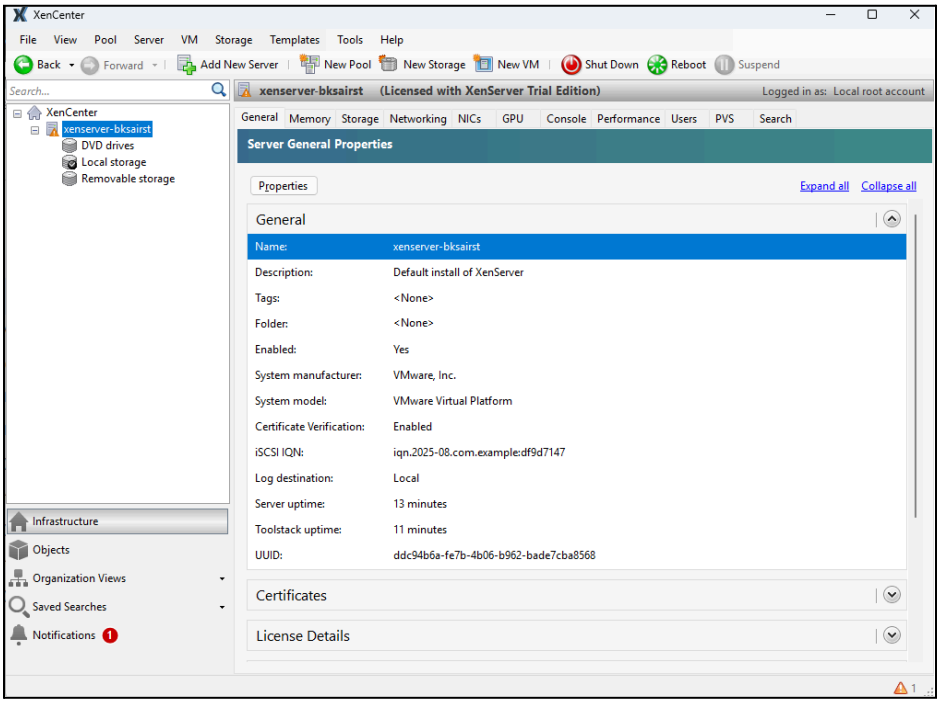
9. Download the xen Center msi file (uploaded in classroom) > keep everything as default > click next



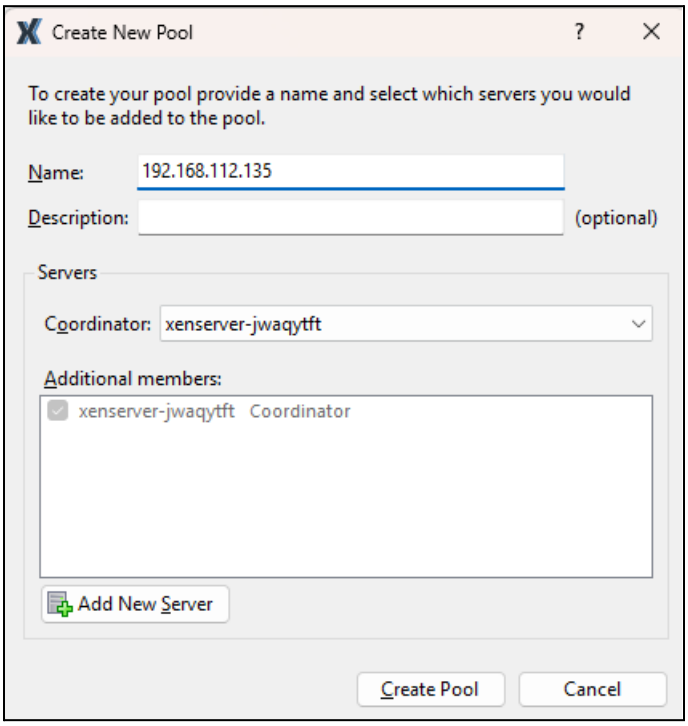
10. Click next > finish > install



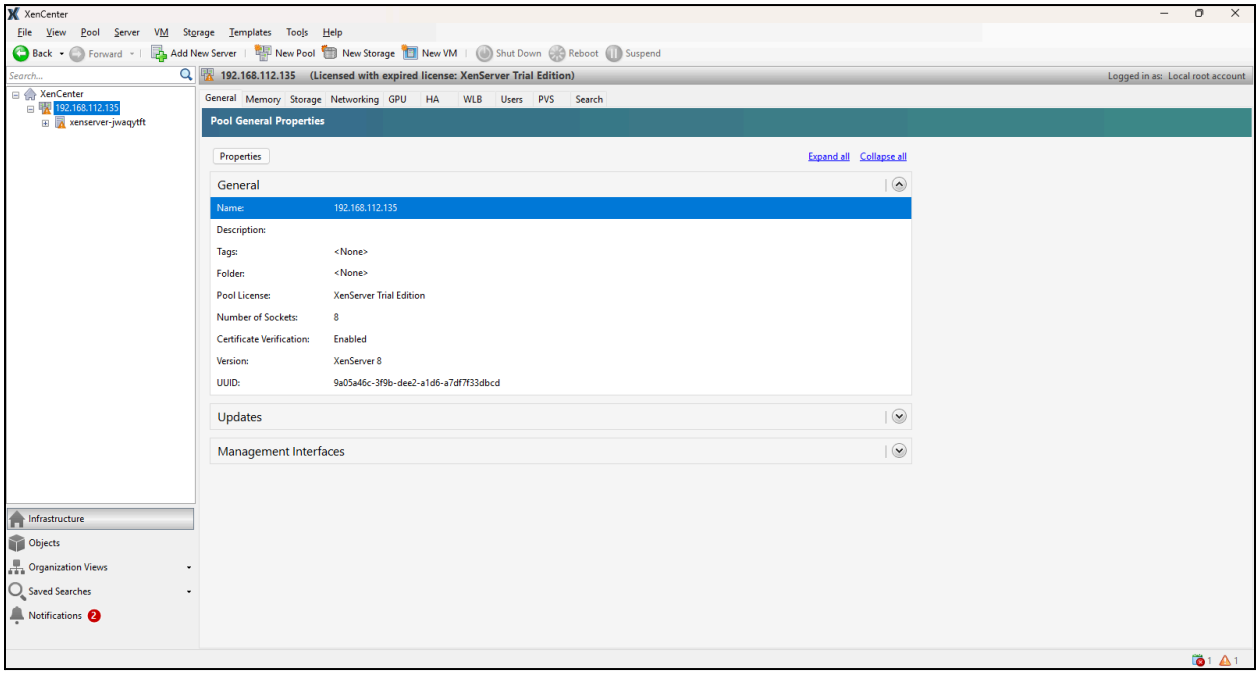
11. Click on add a server > Give your above noted ip address in server > enter the password 1234567 > add > new window will pop up keep it as default > click ok.



12. Go to new pool > create new pool > give the ip address as name > create pool.



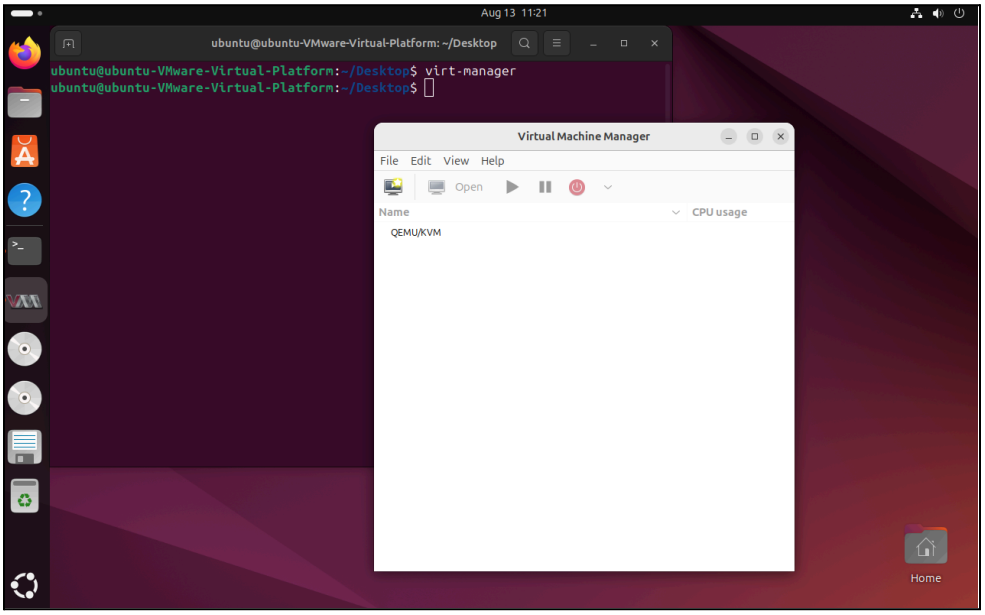
13. New pool is created



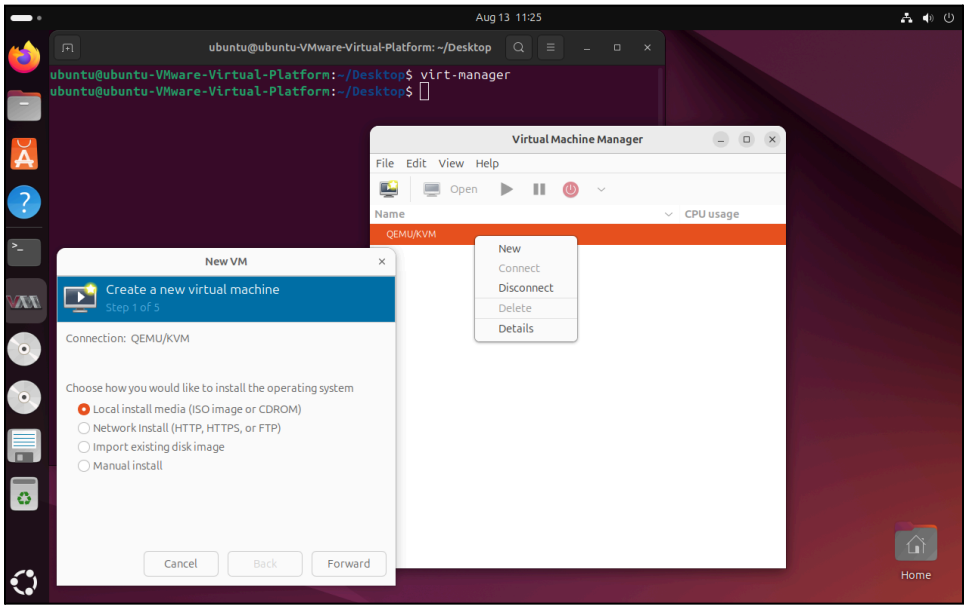
Practical 5 (13-08-25)

Installation and Configuration of virtualization using KVM

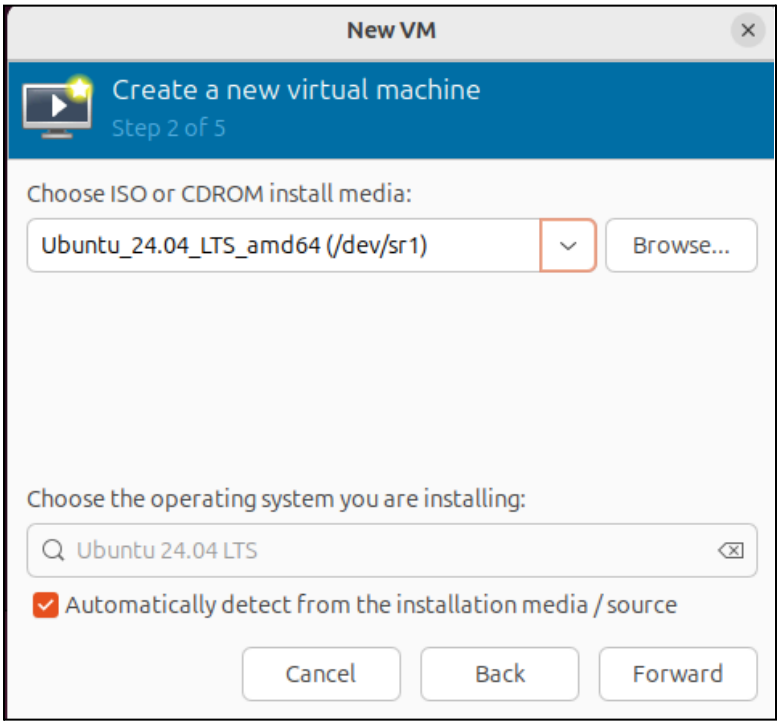
1. Go to VMWare > Open ubuntu VM > open the terminal > in desktop directory type virt-manager.



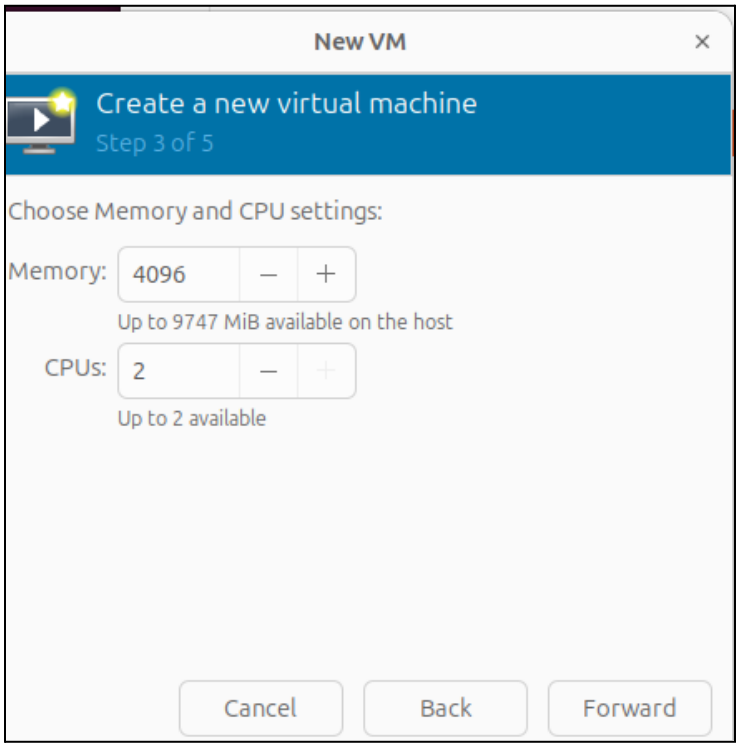
2. Right click on QEMU/KVM > New virtual machine > Select local media >forward.



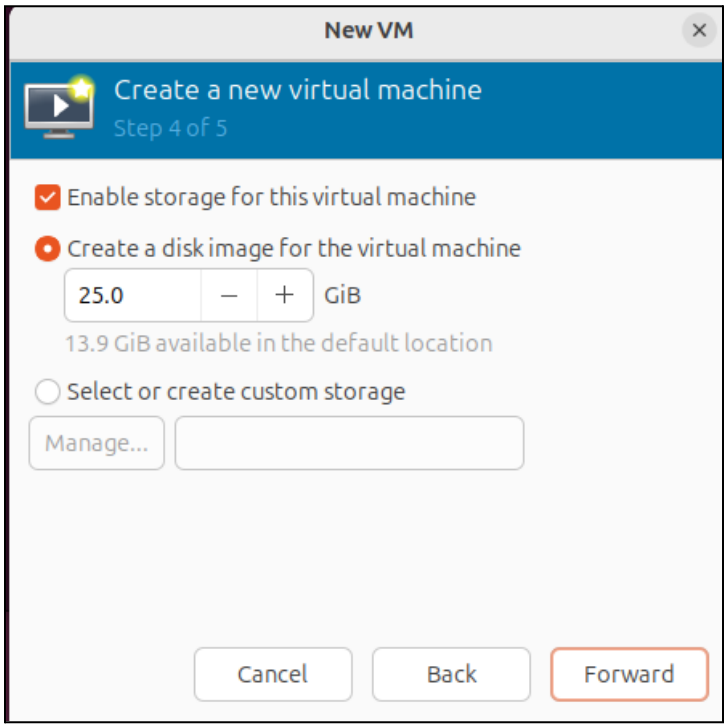
3. Select Ubuntu > forward



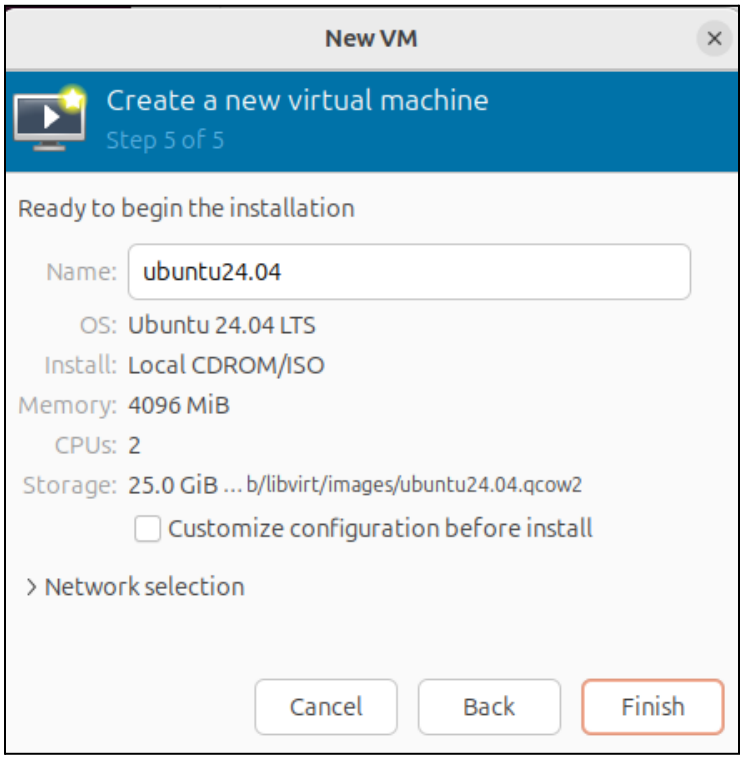
4. Keep everything as default > forward.



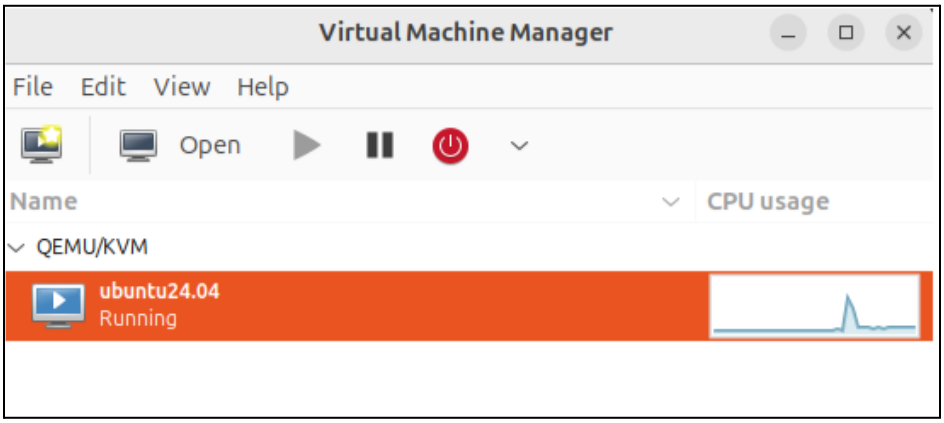
5. Keep everything as default > forward.



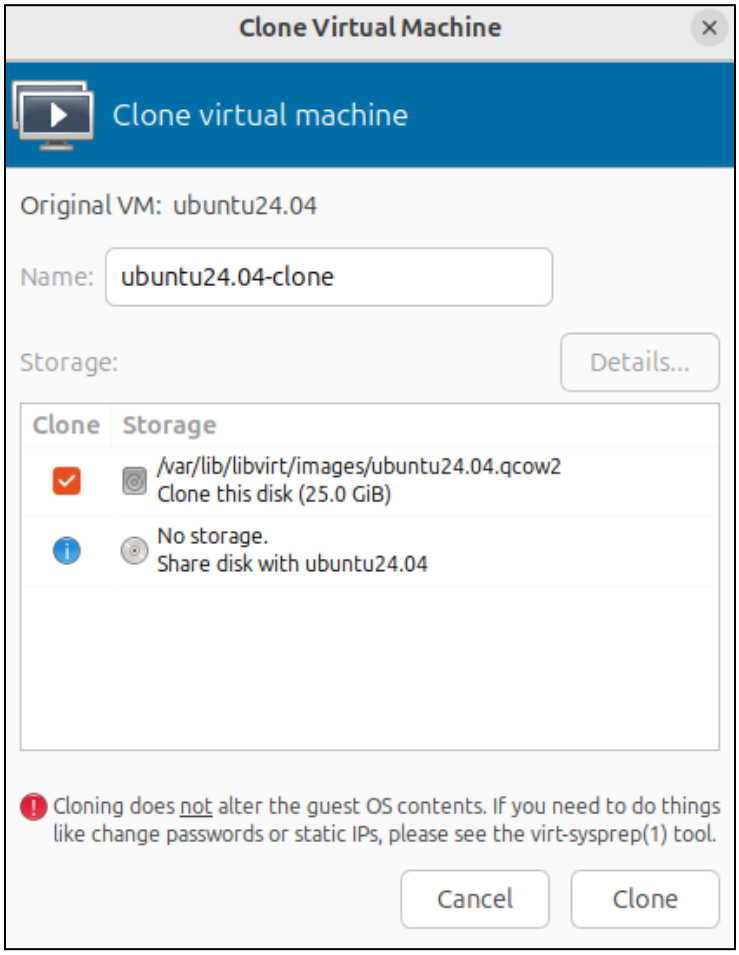
6. Keep everything as default > Finish.



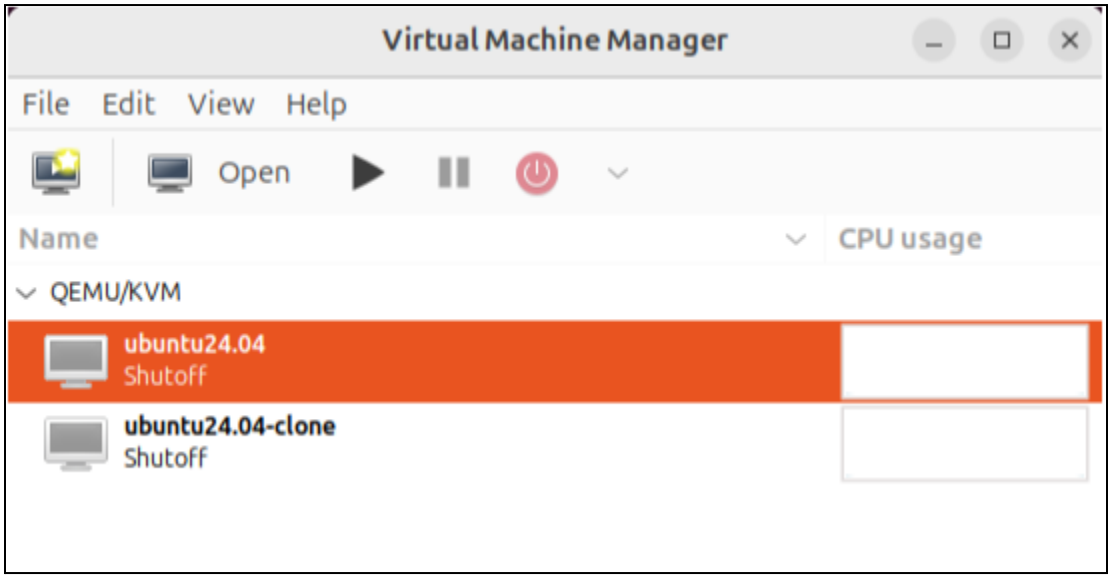
7. After finish > ubuntu vm will be created in ubuntu vm > but ubuntu machine will not start because of disk space > still we can see it running by coming out.



8. Right click on ubuntu vm > force off > right click again > clone > new window will pop up > clone.



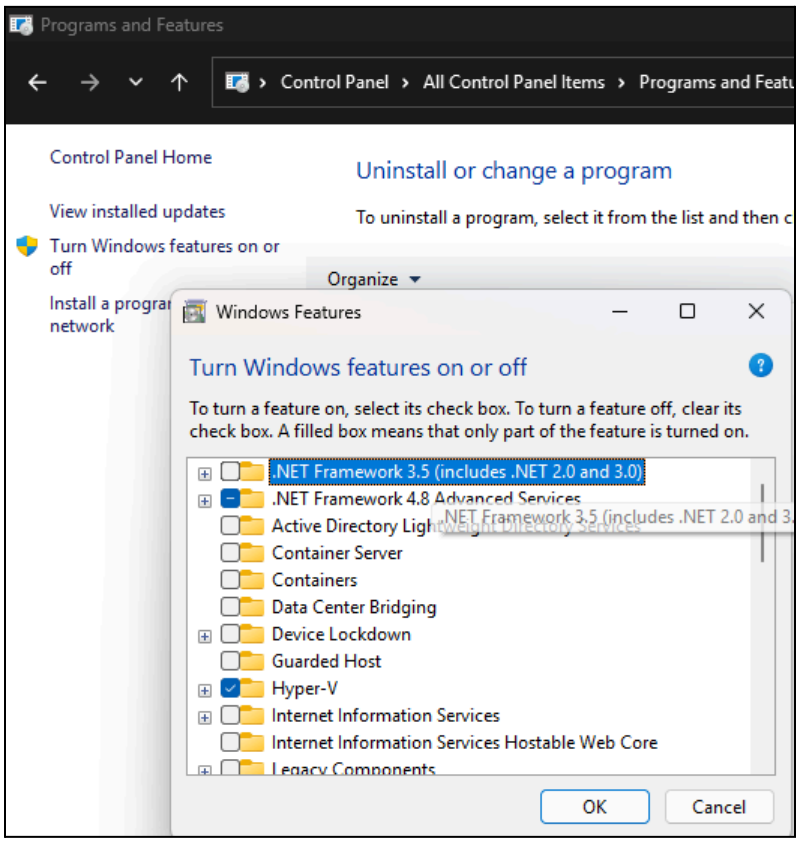
9. The clone of your VM is Created!!



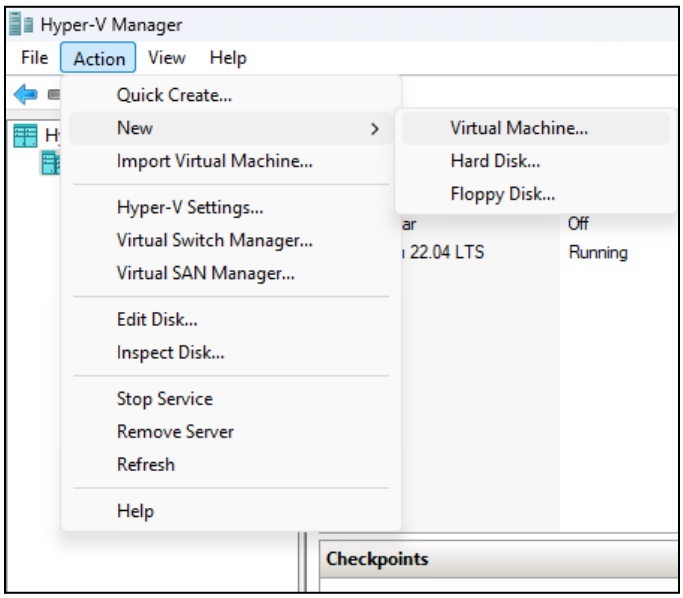
Practical 6 (09-07-25)

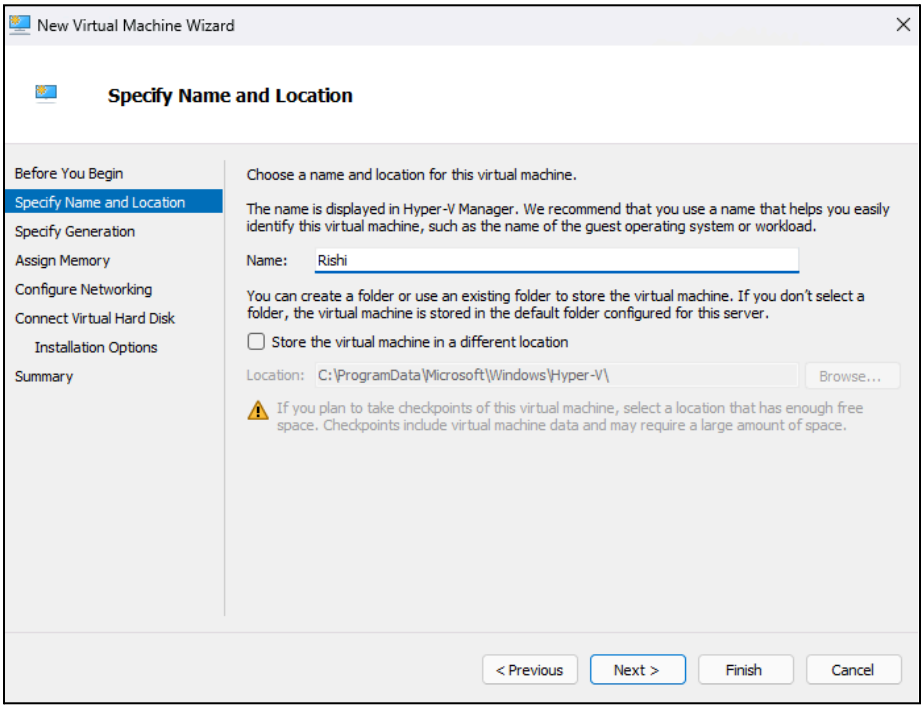
Implement Windows Hyper V virtualization

1. Go to Control Panel > Programs and Features > Turn WIndow features on or off > plus symbol for Hyper V and turn everything on > press ok and restart



2. After turning on > search hyper v manager > go to your pc name > press action > new > virtual machine > click next and give name > next select gen 1 > next > select default switch > change size to 150 gb > finish





Choose the generation of this virtual machine.

- (●) Generation 1

  This virtual machine generation supports 32-bit and 64-bit guest operating systems and provides virtual hardware which has been available in all previous versions of Hyper-V.

- ( ) Generation 2

  This virtual machine generation provides support for newer virtualization features, has UEFI-based firmware, and requires a supported 64-bit guest operating system.

⚠ Once a virtual machine has been created, you cannot change its generation.

New Virtual Machine Wizard

Configure Networking

- Before You Begin
- Specify Name and Location
- Specify Generation
- Assign Memory
- Configure Networking
- Connect Virtual Hard Disk
- Installation Options
- Summary

Each new virtual machine includes a network adapter. You can configure the network adapter to use a virtual switch, or it can remain disconnected.

Connection: Default Switch

< Previous | Next > | Finish | Cancel

Connect Virtual Hard Disk

- Before You Begin
- Specify Name and Location
- Specify Generation
- Assign Memory
- Configure Networking
- Connect Virtual Hard Disk
- Installation Options
- Summary

A virtual machine requires storage so that you can install an operating system. You can specify the storage now or configure it later by modifying the virtual machine's properties.

(●) Create a virtual hard disk

Use this option to create a VHDX dynamically expanding virtual hard disk.

Name: Rishi.vhdx

Location: C:\ProgramData\Microsoft\Windows\Virtual Hard Disks\ Browse...

Size: 150 GB (Maximum: 64 TB)

( ) Use an existing virtual hard disk

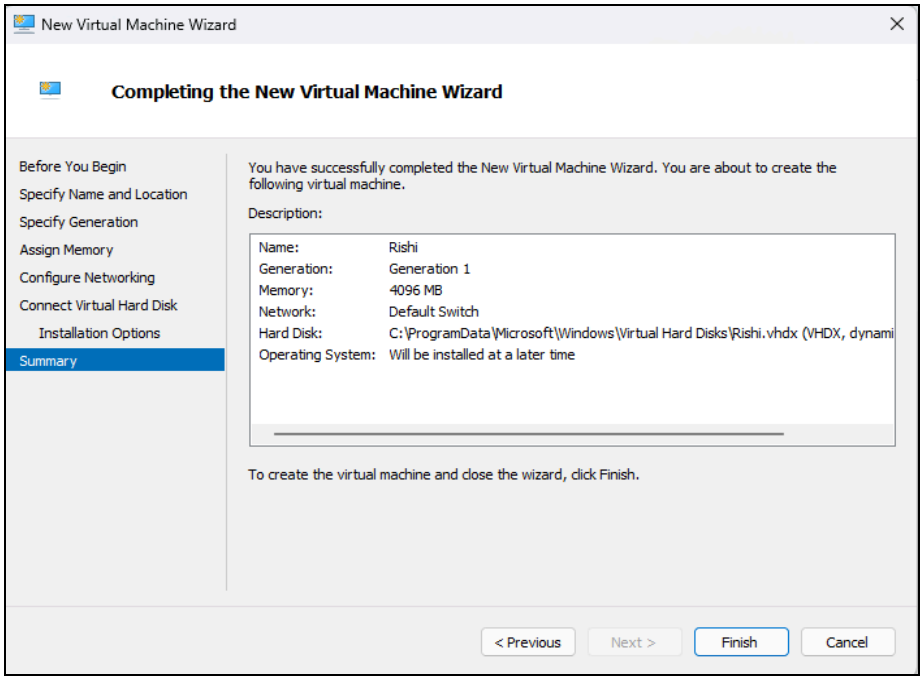
Use this option to attach an existing virtual hard disk, either VHD or VHDX format.

Location: C:\ProgramData\Microsoft\Windows\Virtual Hard Disks\ Browse...

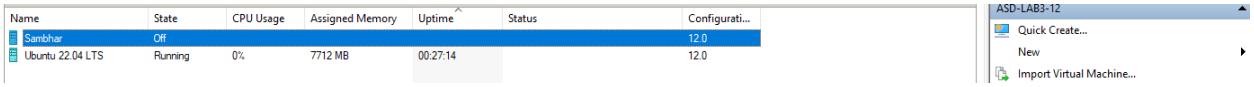
( ) Attach a virtual hard disk later

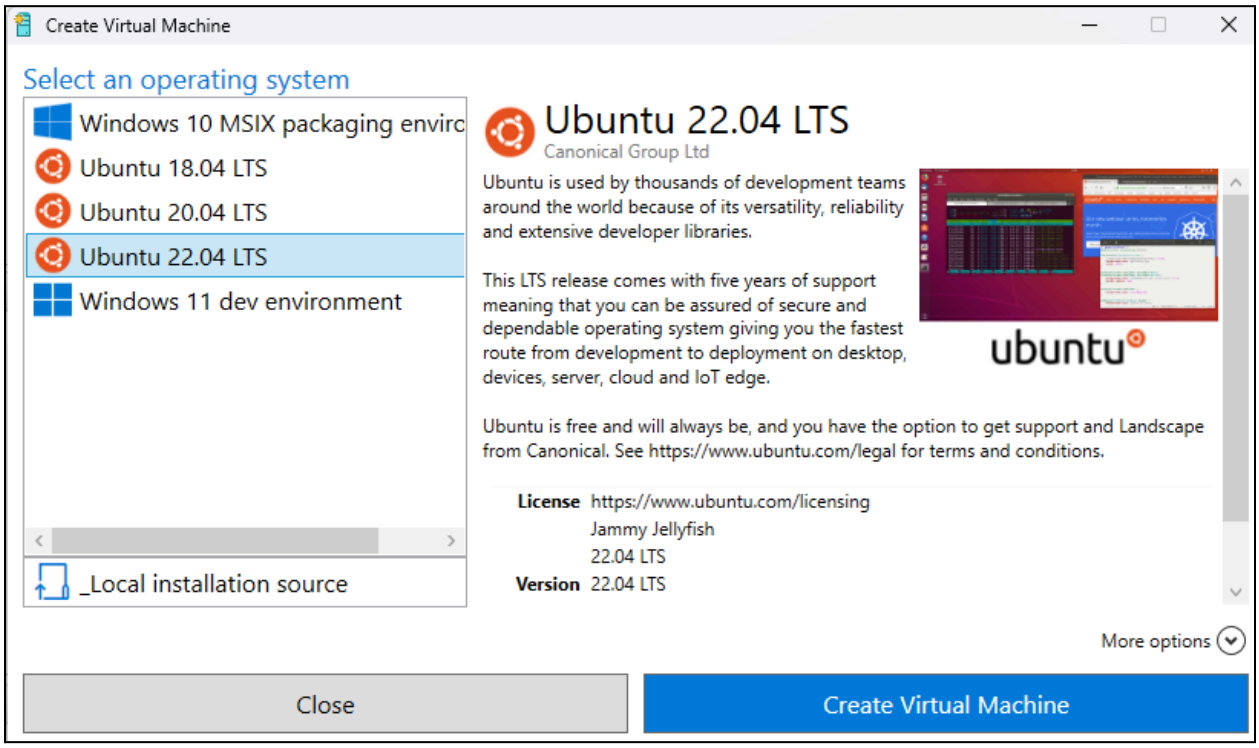
Use this option to skip this step now and attach an existing virtual hard disk later.

< Previous | Next > | Finish | Cancel



3. Select your vm and click quick create > select ubuntu 22.04 and download > set your username and password. Now you have successfully installed a virtual machine.





Practical 7 (06-08-25)

Write a program on Web Feed

Q. Create a registration form and store the details in the database.
Steps:
create folder
npm init -y
npm install express mongoose body-parser
make 2 new files - server.js and index.html
then in atlas create a new cluster, copy your connection string and put it in the code, then connect the user by clicking 'database access' with your username and pass, then copy, and in network access, make a new one and allow from anywhere
then run your program by writing node server.js, there you see the link, ctrl+press and open that link
enter details and submit
then go to cluster and browse collections, there you will see you entry
*now if you want to change collection name, write the name before the ? -> mongodb.net/**prac**?retry,and then do the changes in both the code for accepting, DONE.*

server.js

```
const express = require('express');
const mongoose = require('mongoose');
const path = require('path');

const app = express();
const port = 3000;
const mongoURI =
"mongodb+srv://ruchib_db_user:1o4TPcx4UklvL2y6@cluster0.tk9ezij.mongodb.n
et/?retryWrites=true&w=majority&appName=Cluster0";

// --- MIDDLEWARE ---
// Use the built-in Express middleware to parse form data
app.use(express.urlencoded({ extended: true }));

// --- DATABASE CONNECTION ---
```

```
mongoose.connect(mongoURI)
  .then(() => console.log('✅ MongoDB connected successfully!'))
  .catch(err => console.error('MongoDB connection error:', err));

// --- DATA BLUEPRINT (SCHEMA & MODEL) ---
const studentSchema = new mongoose.Schema({
    name: String,
    age: Number,
    grade: String
});
const Student = mongoose.model('Student', studentSchema);

// --- ROUTES ---
// 1. Serve the HTML form to the user
app.get('/', (req, res) => {
    res.sendFile(path.join(__dirname, 'index.html'));
});

// 2. Handle the form submission
app.post('/addstudent', async (req, res) => {
    try {
        const newStudent = new Student(req.body);
        await newStudent.save();
        res.send('<h1>✅ Success! Student details have been saved.</h1><a
href="/">Go Back</a>');
    } catch (err) {
        console.error("Error saving student:", err);
        res.status(500).send('<h1>❌ Error! Could not save details.</h1>');
    }
});

// --- START SERVER ---
app.listen(port, () => {
    console.log(`🚀 Server is running at http://localhost:${port}`);
});
```

index.html

```
const express = require('express');
const mongoose = require('mongoose');
const path = require('path');

const app = express();
const port = 3000;
const mongoURI =
"mongodb+srv://ruchib_db_user:1o4TPcx4UklvL2y6@cluster0.tk9ezij.mongodb.n
et/?retryWrites=true&w=majority&appName=Cluster0";

// --- MIDDLEWARE ---
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  try {
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        const newStudent = new Student(req.body);
        await newStudent.save();
        res.send('<h1>✅ Success! Student details have been saved.</h1><a
href="/">Go Back</a>');
    } catch (err) {
        console.error("Error saving student:", err);
        res.status(500).send('<h1>❌ Error! Could not save details.</h1>');
    }
});

// --- START SERVER ---
app.listen(port, () => {
    console.log(`🚀 Server is running at http://localhost:${port}`);
});
```
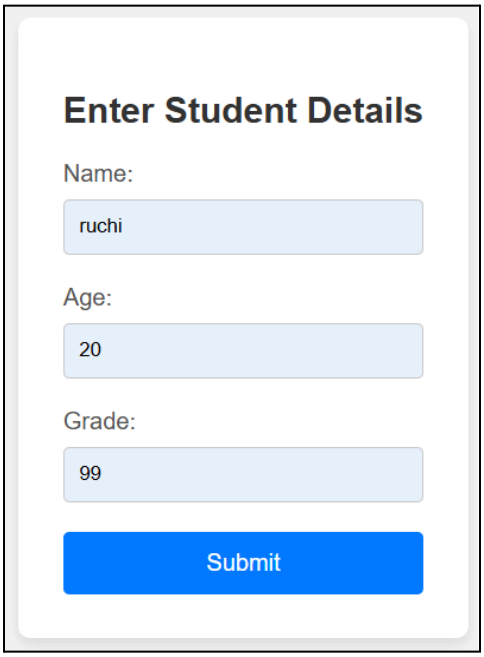
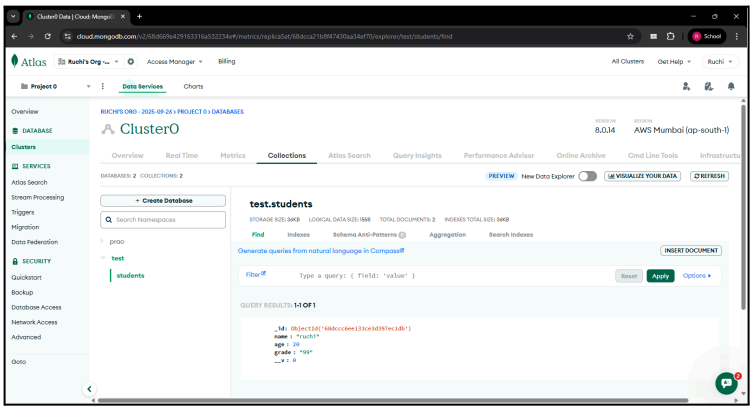

```
PS C:\Users\admin\Documents\prac> node server.js
🚀 Server is running at http://localhost:3000
✅ MongoDB connected successfully!
```





✅ **Success! Student details have been saved.**

Go Back

Practical 8 (09-07-25)

Case Study (Microsoft Azure)

1. What is Azure
->
Microsoft Azure is a cloud computing platform and service created by Microsoft. It provides a wide range of cloud services including computing, analytics, storage, and networking. Users can build, deploy, and manage applications and services through Microsoft-managed data centers.

2.History of Azure
->

- 2008: Announced by Microsoft as “Windows Azure.”
- 2010: Officially launched to the public.
- 2014: Rebranded to “Microsoft Azure” to reflect its expanding service scope beyond just Windows.

Since then, Azure has continuously grown, adding new services and expanding globally.

3.First to fourth generation of Azure
->
Azure generations generally refer to the evolution of its architecture and services:

- 1st Generation (2010 - 2013): Basic PaaS model focusing on Windows-based apps, initially called Windows Azure.
- 2nd Generation (2013 - 2016): Expanded IaaS capabilities, including support for Linux and Virtual Machines.
- 3rd Generation (2016 - 2020): Introduction of AI, big data, IoT, Kubernetes, and DevOps integrations.
- 4th Generation (2020 - Present): Focus on hybrid cloud with Azure Arc, edge computing, serverless, and enhanced security.

4.Industry adoption of Azure

->
Azure is widely adopted across industries such as:

- Healthcare: For secure patient data management and AI-powered diagnostics.
- Finance: For scalable and compliant infrastructure.
- Retail: For e-commerce platforms and analytics.
- Government: For secure cloud migration and digital services.
- Manufacturing: IoT and predictive maintenance.

Many enterprises choose Azure for its hybrid cloud capabilities and integration with Microsoft products.

5.Azure Market place
->
Azure Marketplace is an online store that offers thousands of certified, open-source, and community software applications and services that run on Azure. It enables customers to find, try, and buy software solutions built by Microsoft and its partners.

6.Services of Azure
->
Azure provides over 200 services; key ones include:

- Compute: Virtual Machines, Azure Functions (serverless)
- Storage: Blob Storage, Disk Storage
- Databases: Azure SQL Database, Cosmos DB
- Networking: Virtual Network, Load Balancer, VPN Gateway
- AI and Machine Learning: Azure Cognitive Services, Azure Machine Learning
- Analytics: Azure Synapse Analytics, HDInsight
- Identity: Azure Active Directory

7.Additional Azure Services

->

- Azure DevOps: CI/CD pipelines and developer tools.
- Azure IoT Hub: Connect, monitor, and manage IoT assets.
- Azure Security Center: Unified security management.
- Azure Arc: Manage resources across hybrid and multi-cloud environments.
- Azure Cognitive Search: AI-powered search as a service.

8. Types of clouds

->

- Public Cloud: Services delivered over the internet by third-party providers (e.g., Azure, AWS, Google Cloud).
- Private Cloud: Cloud infrastructure operated solely for one organization.
- Hybrid Cloud: Combination of public and private clouds, allowing data and apps to be shared between them.
- Multi-Cloud: Use of multiple cloud services from different providers.

9.pros and cons of Azure

->

<u>Pros:</u>

- Wide range of integrated services.
- Strong hybrid cloud support.
- Global data centers with high availability.
- Seamless integration with Microsoft tools.
- Enterprise-grade security and compliance.

<u>Cons:</u>

- Pricing complexity can be confusing.
- Steep learning curve for beginners.
- Some users report slower customer support.
- Occasional service outages impacting availability.

Practical 9 (09-07-25)

Case Study (Google Cloud Platform)

1. What is Google cloud
->
Google Cloud (Google Cloud Platform – GCP) is Google's suite of public cloud computing services, offering infrastructure (IaaS), platform (PaaS), and managed services such as compute, storage, networking, big data analytics, and AI/ML tools. It runs on the same global infrastructure that powers Google Search, Gmail, YouTube, and other Google products.

2.History of Google cloud
->
- 2008: Google App Engine launched as a PaaS service.
- 2010: Introduction of Cloud Storage, BigQuery, and Prediction API.
- 2012: Google Compute Engine (IaaS) announced in preview, with general availability in 2013.
- 2014–2015: Launch of Kubernetes (open-sourced), Google Kubernetes Engine (GKE), Pub/Sub, Dataflow, and Cloud Functions.
- 2019–2023: Expansion into Anthos (multi-cloud), Cloud Run, Vertex AI, and generative AI integrations.

3.Generation of Google cloud
->
1. PaaS-first era – App Engine launch.
2. IaaS expansion – Compute Engine and Cloud Storage.
3. Data and analytics leadership – BigQuery, Dataflow, Pub/Sub.
4. Container-native era – Kubernetes, GKE, Anthos.
5. AI/ML-driven evolution – Vertex AI, AutoML, Gemini AI models.

4.Industry adoption of Google cloud
->
Google Cloud is adopted across industries like financial services, retail, healthcare, manufacturing, and government. It is popular in analytics, AI, and

data-heavy workloads. In India, banks like HDFC use cloud-native solutions for scalability, while technology companies use it for multi-cloud and AI transformation.

5.Google cloud Market place

->

Google Cloud Marketplace is a platform where customers can find, purchase, and deploy third-party and partner software solutions directly into their Google Cloud environment. These solutions are pre-integrated and optimized for GCP.

6.Services of Google cloud

->

- Compute: Compute Engine, App Engine, Kubernetes Engine, Cloud Run.
- Storage & Databases: Cloud Storage, Cloud SQL, Bigtable, Firestore, Spanner.
- Data & Analytics: BigQuery, Dataflow, Dataproc, Pub/Sub, Looker Studio.
- AI & ML: Vertex AI, AutoML, Tensor Processing Units (TPUs).
- Networking & Security: Virtual Private Cloud (VPC), Cloud DNS, IAM, Cloud Load Balancing.
- Developer Tools: Cloud Build, Cloud Functions, Cloud Monitoring, Cloud Logging.

7.Additional Google cloud Services

->

- Anthos – Multi-cloud and hybrid Kubernetes management.
- Apigee – API management platform.
- IoT Core – Internet of Things device management.
- Cloud Armor – Security for apps and websites.
- Operations Suite – Advanced monitoring, logging, and tracing.

8.Types of Google cloud

->

- Infrastructure as a Service (IaaS) – Compute Engine, Persistent Disks.
- Platform as a Service (PaaS) – App Engine, Cloud Functions.
- Serverless Computing – Cloud Run, BigQuery serverless analytics.
- Managed AI/ML – Vertex AI, AutoML.
- Database Services – Spanner, Bigtable, Cloud SQL.

- Networking & Security Services – VPC, IAM, Key Management.

9.pros and cons of Google cloud
->
Pros:

- High-performance global infrastructure.
- Strong AI/ML and big data capabilities.
- Robust security and compliance features.
- Integrated developer tools and APIs.

Cons:

- Can be costlier than some competitors for similar workloads.
- Smaller market share compared to AWS and Azure in some regions.
- Learning curve for beginners.
- Complex IAM and security setup for large organizations.

10. Case study Google cloud 1.Reliance 2.HDFC Bank 3. L& T 4.TCS (Any two).
->
HDFC Bank
HDFC Bank adopted cloud-native solutions to handle high transaction volumes without affecting core banking systems. They used Google Cloud tools for scalability, performance, and better customer engagement.

Tata Consultancy Services (TCS)
TCS uses Google Cloud to deliver AI-powered analytics and digital transformation services to its clients. They leverage Google Cloud's big data and machine learning capabilities to optimize business operations and improve customer experiences.

Practical 10 (13-08-25)

Case study on cloud data privacy and security issues, jurisdictional issues

Executive Summary

After reviewing policy documents (EU GDPR, India DPDP Act, U.S. CLOUD Act) and Microsoft's own transparency reports, it became clear that organizations moving sensitive workloads to Azure face two intertwined challenges:

- Jurisdictional exposure – data stored in one country can be subject to foreign lawful access via the CSP.
- Operational risk – accidental cross-border replication, weak key custody, and over-permissive identities.

Outcome of the implementation: By applying region-locked policies, customer-managed keys in Azure Dedicated HSM, tokenization for analytics, and confidential computing, the architecture reduced cross-border transfers by over 60%, improved incident detection speed, and ensured defensibility against contested legal requests.

Problem Context

Primary goals:

- Keep regulated personal data within the approved region(s).
- Maintain compliance with local privacy laws and contractual obligations.
- Preserve analytics and business intelligence capabilities without leaking raw identifiers.

Risks identified from reports:

- CLOUD Act scenarios where CSP could be compelled to hand over data.
- DPDP Act potential blacklisting of transfer destinations.
- Azure services with default global replication settings.

Jurisdiction-Aware Azure Architecture

1. Region Residency & Policy Enforcement
   - Azure Policy deny assignments for all non-approved regions.
   - Resource groups bound to India West / India South (or other compliance-approved regions).
   - Block GRS and multi-region replication by default.
   - Private Link used for Storage, Synapse, Key Vault — no public endpoints.

2. Key Custody & Cryptography
   - Customer-managed keys (CMK) stored in Azure Dedicated HSM or Managed HSM.
   - Split-key design: decryption requires quorum-based approval.
   - Key release gated by Azure Attestation — only attested Confidential VMs or approved workloads can receive keys.

3. Privacy-Preserving Analytics
   - Tokenization service deployed on Azure Kubernetes Service (AKS).
   - Analytics executed in Azure Databricks or Azure Synapse on pseudonymized datasets.
   - Highly sensitive joins run in Confidential Computing VMs (DCasv5/ECasv5).
   - Non-production vendors receive synthetic datasets only.

4. Identity & Access
   - Unified identity via Microsoft Entra ID.
   - Managed Identities for services; removal of long-lived secrets.
   - Privileged Identity Management (PIM) for just-in-time admin roles.
   - Attribute-based access using Microsoft Purview tags.

5. Observability & Assurance
   - Centralized logging in Microsoft Sentinel (India workspace) with immutable retention.

- Customer Lockbox for Azure enabled to review and approve any Microsoft engineer data access.
- Monthly reconciliation of Lockbox events with Sentinel logs.

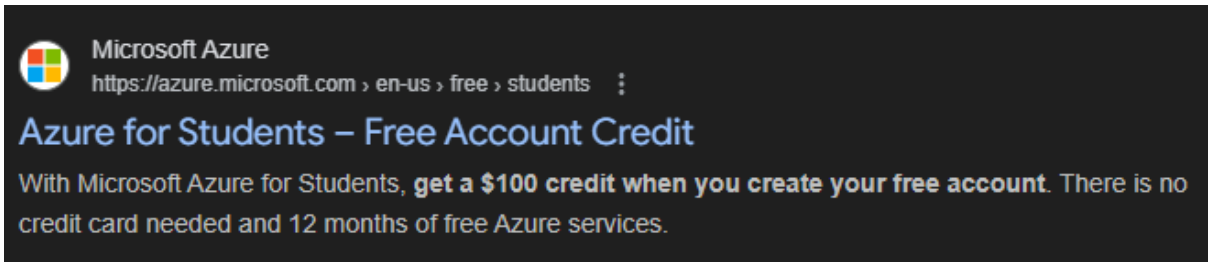
Conclusion

In conclusion, Azure's global infrastructure offers scalability and performance, but data privacy and jurisdictional compliance remain critical concerns. By combining legal awareness with technical measures—such as region restrictions, customer-managed encryption keys, confidential computing, and strict access governance—organizations can significantly reduce the risks of cross-border data exposure and unauthorized access. Ultimately, maintaining compliance in Azure is not just about using its security features, but about implementing them within a clear, enforceable governance framework aligned to the applicable laws of each jurisdiction.

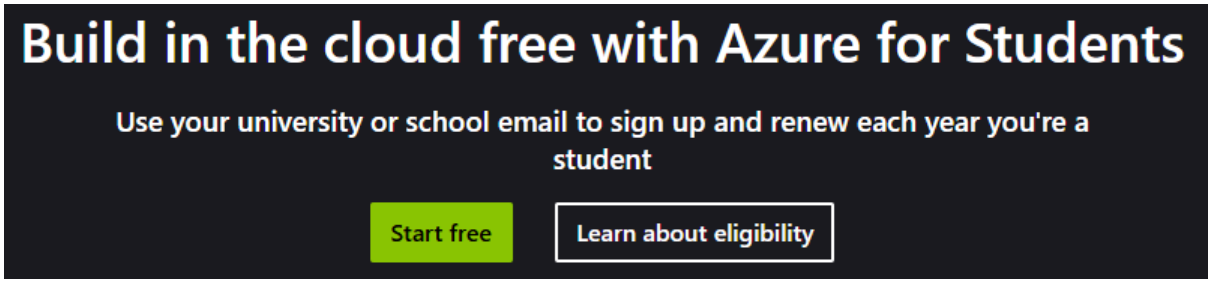
Practical 11 (01-10-25)

Working of Software as a service

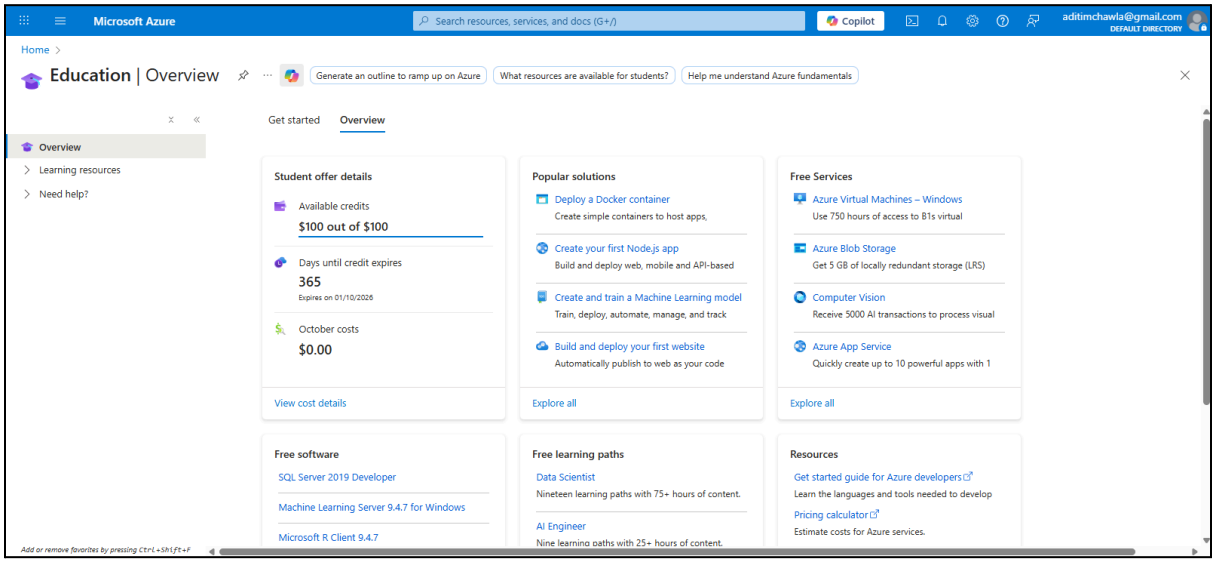
1. Open the first website that appears on Google for Microsoft Azure for Students



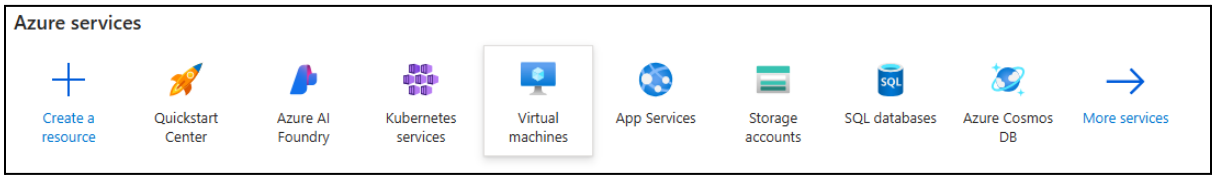
2. Click on Start Free at the top of the website



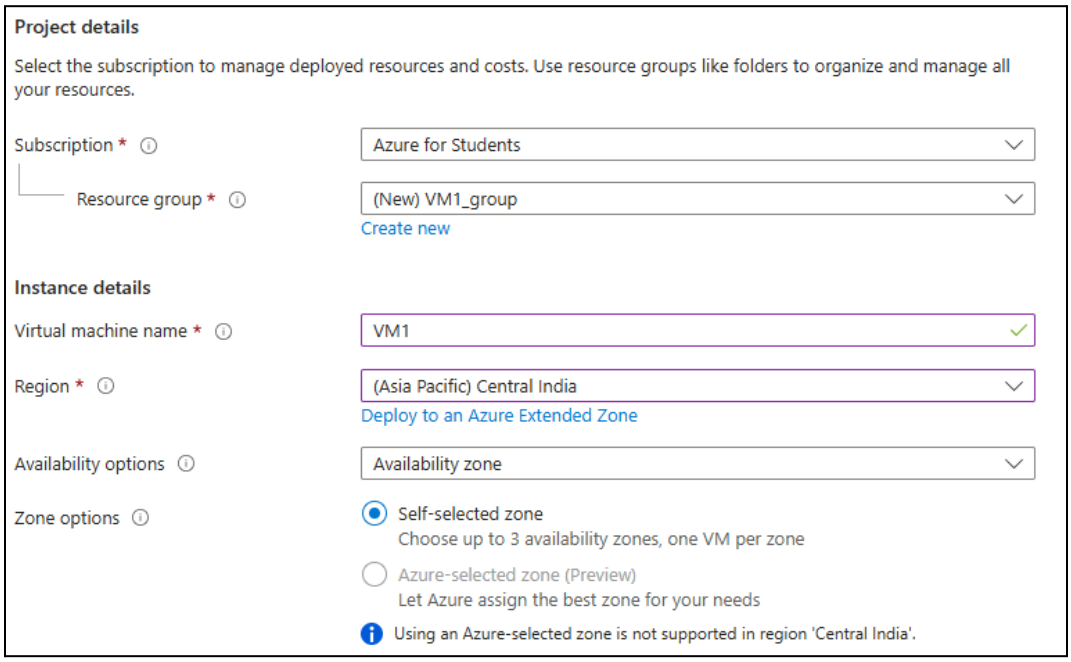
3. Complete login details and home page will open

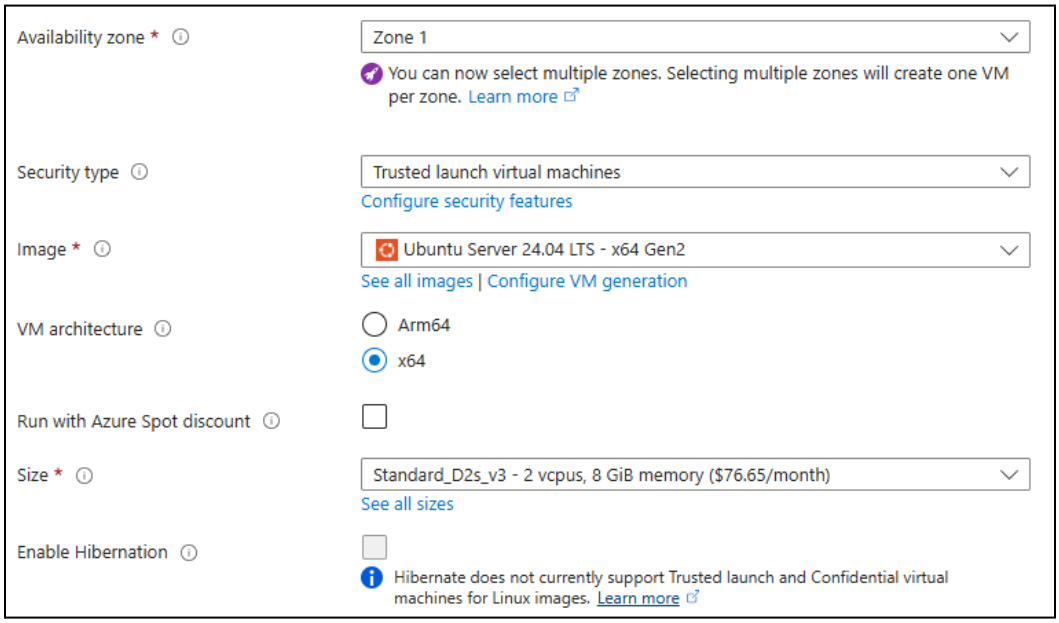


4. Click on Get Started and click Virtual Machines

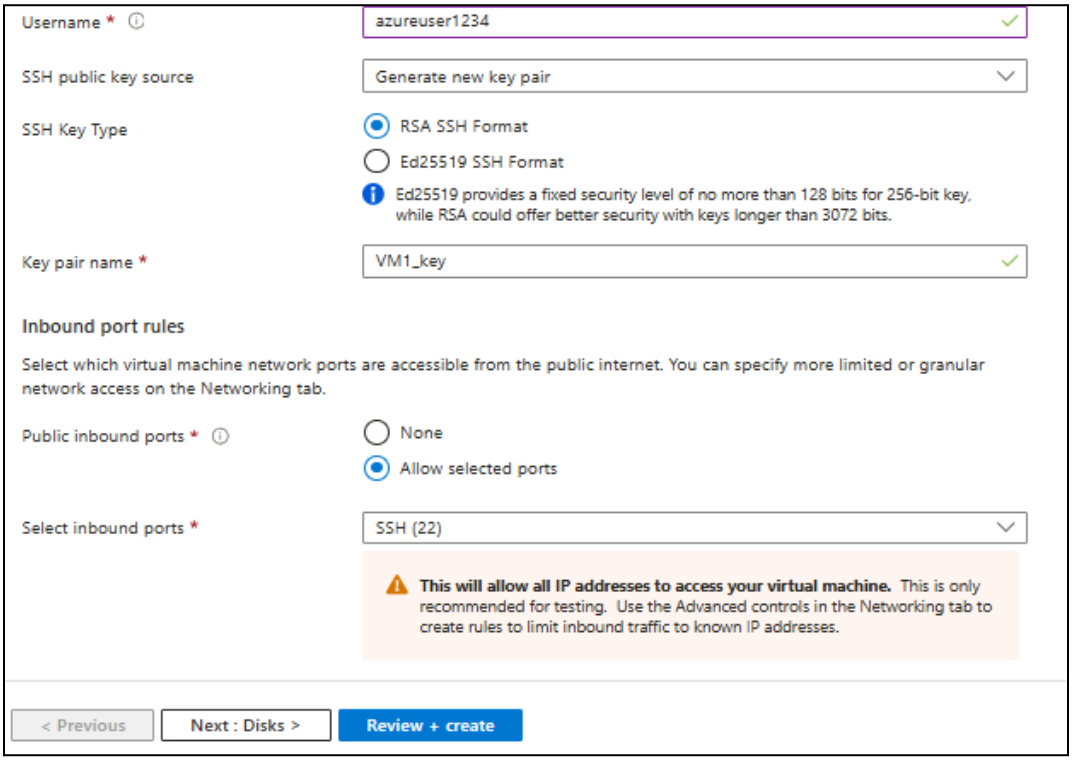


5. Fill in the details

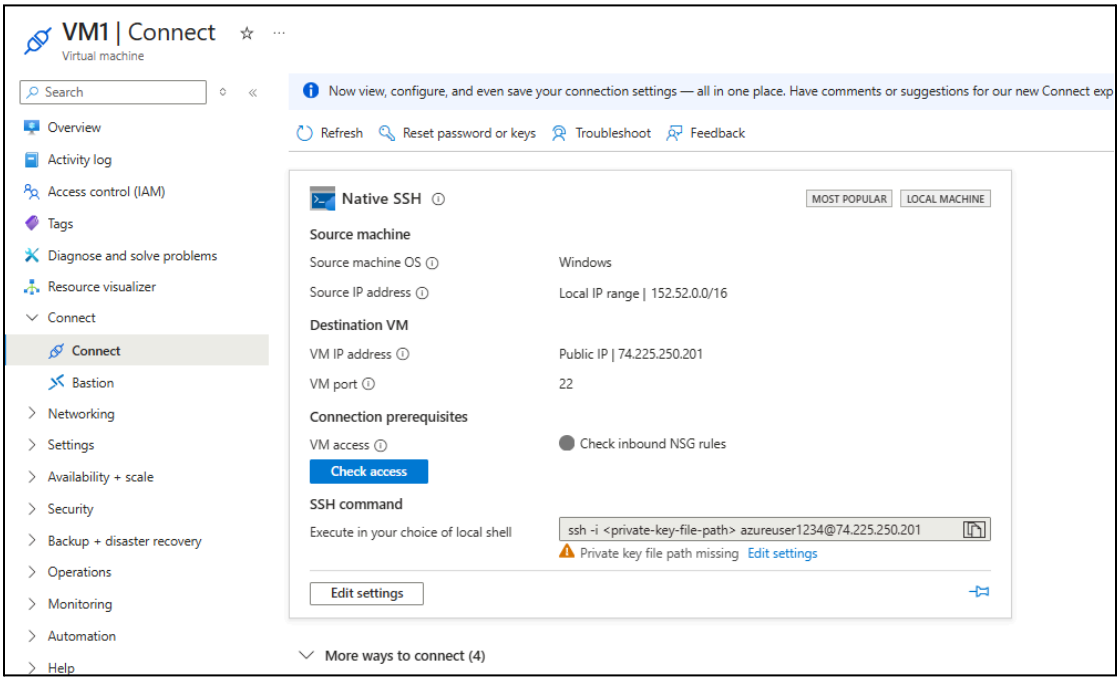




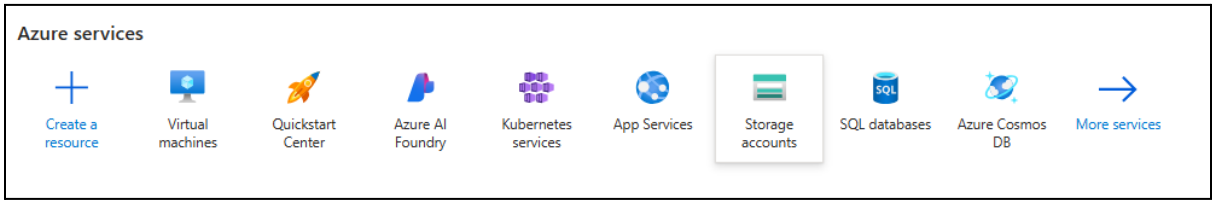
6. Click on Review and Create



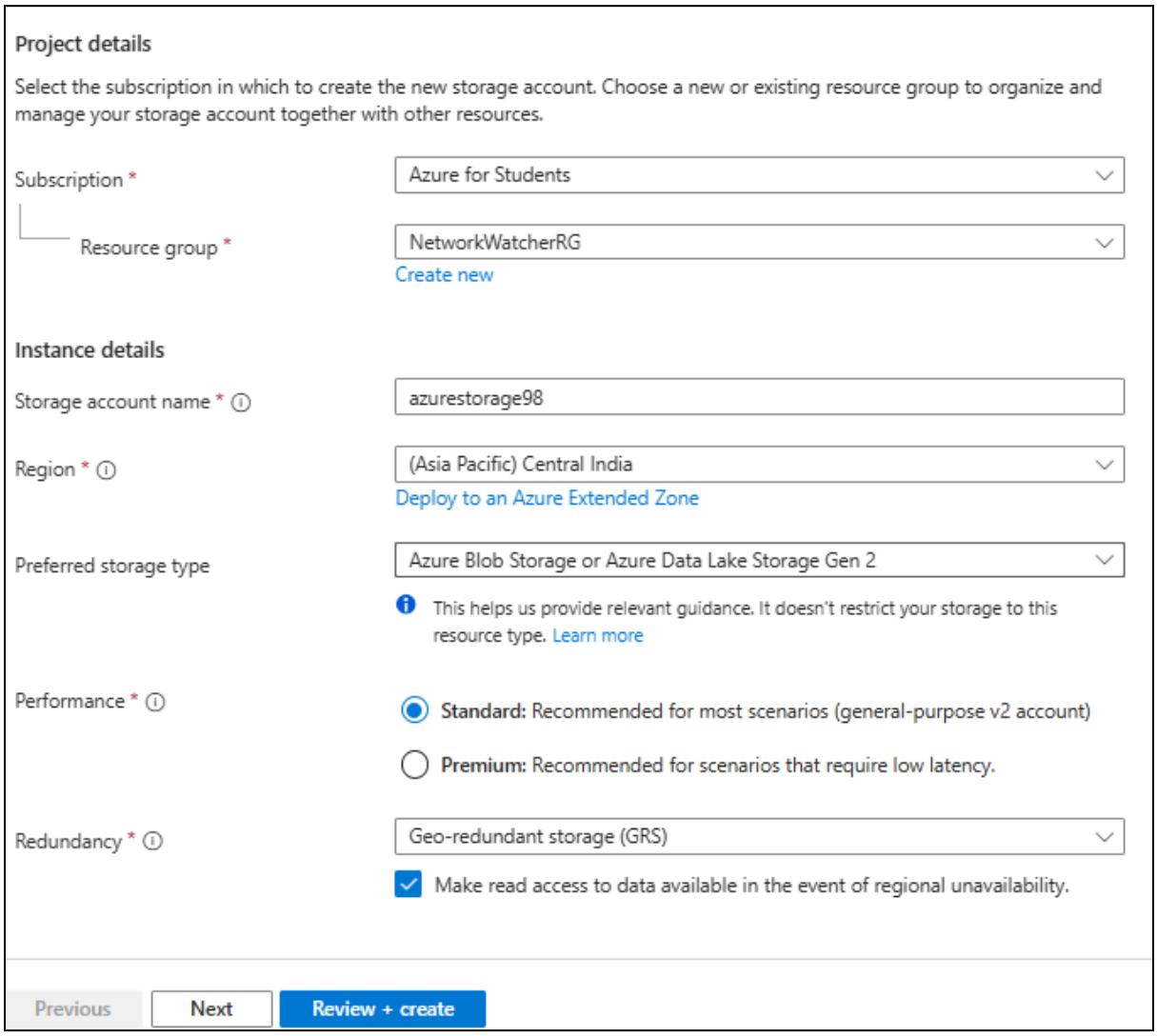
7. VM1 has to be connected to Native SSH



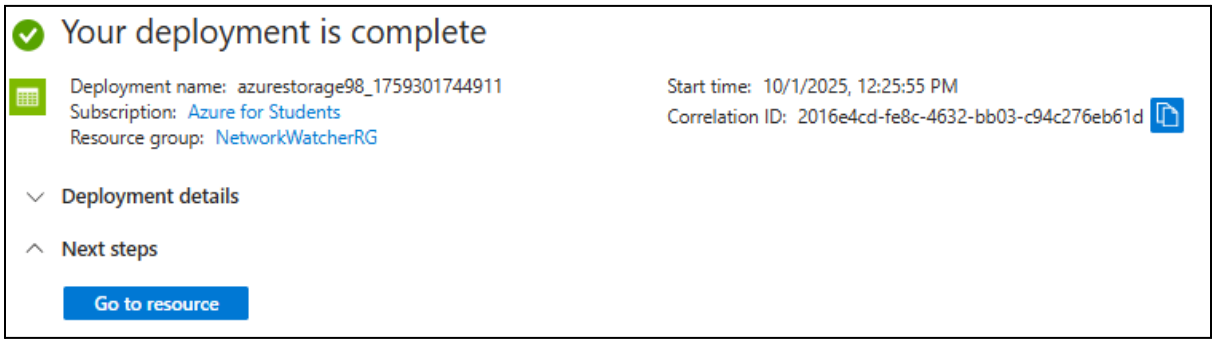
8. Go on Home Page and click on Create Storage



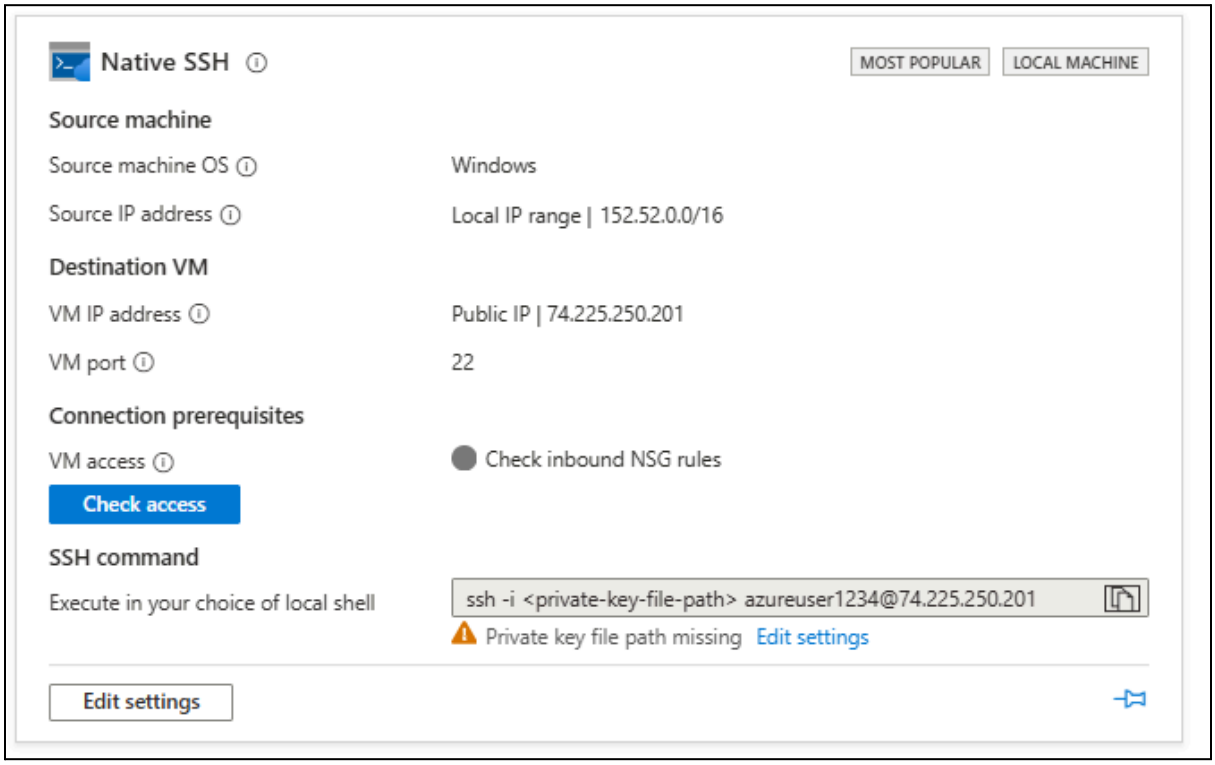
9. Fill in details and click Review + Create



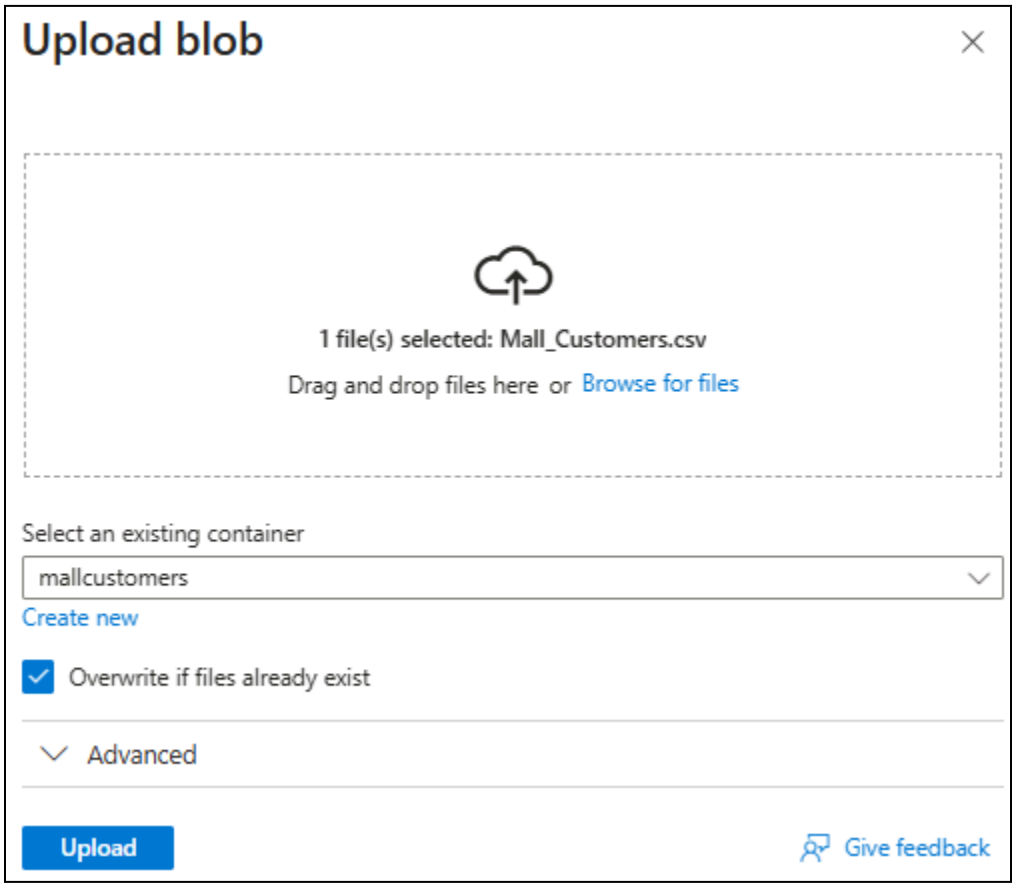
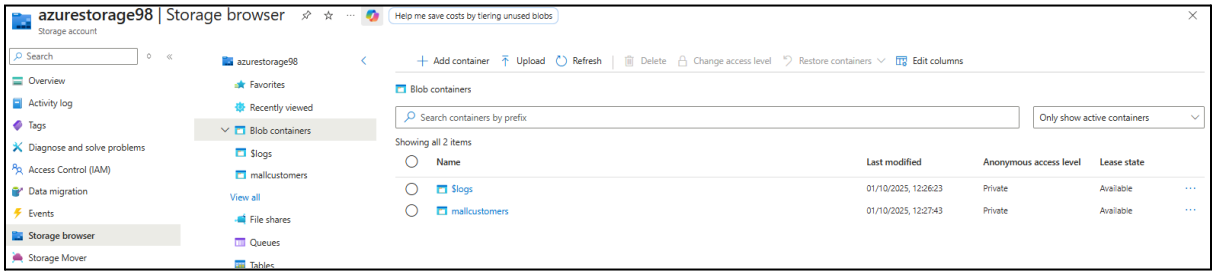
10. Once deployment is complete, click on Go To Resource



11. Azure Storage has been created



12. Upload an excel sheet or any format file



13. The uploaded file can be seen in the storage browser